AD 2. AERODROMES

VOBL AD 2.1 AERODROME LOCATION INDICATOR AND NAME

VOBL - KEMPEGOWDA INTERNATIONAL AIRPORT, BENGALURU

VOBL AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

1	Aerodrome reference point coordinates and its site	131156N 0774220E 15 M East of Control TWR	
2	Direction and distance of aerodrome reference point from the center of the city or town which the aerodrome serves	North-East 30 KM from City Railway station	
3	Aerodrome elevation and reference temperature	3003 FT / 36.6 DEG C	
4	Magnetic variation, date of information and annual change	1.83 DEG W (2010) /0.033 DEG E	
5	Name of aerodrome operator, address, telephone, telefax, e-mail address, AFS address, website (if available)	Administrative Block, Bravo-1, Kempegowda International Airport, Bengaluru, (India) PIN-560300,	
		Telephone:	(91) 8066782222 (91) 8066782204
		Fax:	(91) 8066783366
		AFS:	----
		Email:	aoccmanager@bialairport.com
6	Types of traffic permitted (IFR/VFR)	IFR/VFR	
7	Remarks	1. Elevation in EGM-08	

VOBL AD 2.3 OPERATIONAL HOURS

1	Aerodrome Operator	MON-FRI 0300-1200 UTC (0830-1730 IST)
2	Custom and immigration	H24
3	Health and sanitation	H24
4	AIS briefing office	H24
5	ATS reporting office (ARO)	H24
6	MET Briefing office	H24
7	Air Traffic Service	H24
8	Fuelling	H24
9	Handling	H24
10	Security	H24
11	De-icing	NIL

12	Remarks	1.The ATS approved hourly RWY traffic handling capacity as follows: a. RWY09R/27L and RWY09L/27R Single RWY mode Maximum number of Departures only- 33 Maximum number of Arrivals only - 33 Maximum number of Arrivals and Departures - 42 b. RWY09R/27L and RWY09L/27R in Seggregated mode Maximum number of Departures in Departure RWY- 35 Maximum number of Arrivals in Arrival RWY - 35 Maximum number of Arrivals and Departures- 60 NOTE: One ATM of ATR/CODE E/CODE F will be cosidered as two ATMS

VOBL AD 2.4 HANDLING SERVICES AND FACILITIES

1	Cargo-handling facilities	AVBL
2	Fuel and Oil types	JET A1
3	Fuelling facilities and capacity	Fuel Hydrants, Fuel dispensers and Fuel bowsers.
4	De-icing facilities	NIL
5	Hangar space for visiting aircraft	NIL
6	Repair facilities for visiting aircraft	NIL
7	Remarks	Bridge Mounted Equipment is mandatory for the aircraft allocated on Contact Stands.

VOBL AD 2.5 PASSENGER FACILITIES

1	Hotel(s) at or in the vicinity of aerodrome	At Airport in front of Paasenger Terminal Building and in the city.
2	Restaurant(s) at or in the vicinity of aerodrome	At Aerodrome and in the city.
3	Transportation possibilities	Taxies, coaches & buses
4	Medical Facilities	At Airport
5	Bank and post office at or in the vicinity of aerodrome	Banks: At Airport Post office: At Airport
6	Tourist office	At Airport
7	Remarks	NIL

VOBL AD 2.6 RESCUE AND FIRE FIGHTING SERVICES

1	Aerodrome category for fire fighting	Within ATS HR: CAT-10
2	Rescue equipment	AVBL as per the Category

3	Capability for removal of disabled aircraft	1. Bangalore airport has Disabled Aircraft Recovery kit. The critical aircraft with regards to removal capability for the airport is B 747-800 Code F (Excluding A-380). 2. Primary responsibility for removal of disabled aircraft rests with the concerned airline. 3. The airlines may contact 'Aerodrome Coordinator for Disabled Aircraft Removal Operations' for coordinating with BLR for having Disabled Aircraft Recovery Kit. 4. Contact Details of Aerodrome Coordinator for Disabled Aircraft Removal Operations: Chief Operating Officer, BIAL + 91 9820609714 girish.nair@bialairport.com
4	Remarks	Aerodrome coordinator: COO: +91-80-66782090

VOBL AD 2.7 SEASONAL AVAILABILITY CLEARING

1	Type(s) of clearing equipment	NIL
2	Clearance priorities	NIL
3	Remarks	NIL

VOBL AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS/POSITIONS DATA

1	Designation, surface and strength of aprons	01. Designation: Stand V68 to 38, 59 to 89 Surface: Concrete Strength:PCR 1143/R/B/W/T 02. Designation: Stand 39 to 49, 501 to 510, 51W to 58 Surface: Concrete Strength: PCR 1143/R/B/W/T 03. Designation: Stand 90 to 123, V36 TO V63 Surface: Concrete Strength: PCR 1143/R/B/W/T 04. Designation: Stand 69A to 86A Surface: Concrete Strength: PCR 828/R/B/W/T
2	Designation, width, surface and strength of taxiways	Refer AD 2.23 for details
3	Location and elevation of altimeter checkpoints	Location:All Stands Elevation: Refer AD 2.23 for details
4	Location of VOR checkpoints	Runway 09L/27R – RWY holding positions at Taxiways A1 and A11 Runway 09R/27L - Not Available
5	Position of INS checkpoints	NIL
6	Remarks	Refer AD 2.23 for additional information

VOBL AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

1	Use of aircraft stand identification signs, taxiway guidelines and visual docking/parking guidance system at aircraft stands	01. Illuminated stand identification boards displaying stand number and stand co-ordinates are available for stands V36, V37, V38, V39, V40, V41, V42, V43, V44, V45, V46, V47, V48, V49, V50, V51, V52, V53, V54, V55, V56, V57, V58, V59, V60, V61, V62, V63, V68, V69, V70, 01,02, 03, 04, 05, 06, 07, 08, 09, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 22, 23,24, 25, 26, 27, 28, 29,30,31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 45, 47, 49, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 51W, 51, 52W, 52, 53W, 53, 54W, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67,68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81,82, 83, 84, 85, 86, 87, 88, 89, 69A, 71A, 72A, 74A, 75A, 77A, 78A, 80A, 81A, 83A, 84A, 86A, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123. 02. Taxiing guidance signs are provided on all TWY intersections and straight sections. 03. Advanced Visual Docking Guidance System installed on stands V36, V37, V38, V39, V40, V41, V42, V43, V44, V45, V46, V47, V48, V49, V50, V51, V52, V53, V54, V55, V56, V57, V58, V59, V60, V61, V62, V63, V68, V69, V70, 02, 03, 05, 06, 08, 09, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 22, 23, 25, 26, 28, 29, 31, 32, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 106, 107, 108, 111, 112, 113, 115, 116, 117, 118, 120, 121 and 123. Aircraft to follow AVDGS for parking. 04. Aircraft Stand Manoeuvring Guidance Lights are available for stands V36, V37, V39, V40, V42, V43, V45, V46, V48, V49, V51, V52, V54, V55, V57, V60, V62, V63, 2, 3, 5, 6, 8, 9, 11 12, 13, 15, 16, 17, 19, 20, 23, 26, 29, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106. 05. Marshalling guidance available for parking in assigned stands through Authorized Ground Handling agents (GHA)

2	Runway and taxiway markings and lights	RWY Markings: Runway 09L/27R - Runway Designation, Runway Threshold, Runway Touchdown zone, Runway Centre line, Runway Aiming Point, Runway Side stripe markings. Runway 09R/27L - Runway Designation, Runway Threshold, Runway Touchdown zone, Runway Centreline, Runway Aiming Point, Runway Side stripe markings. Lights: Runway 09L/27R – Runway Threshold & Wing bar, Runway Edge, Runway Centreline & Runway End lights. Runway 09R/27L – Runway Threshold & Wing bar, Runway Edge, Runway Touchdown zone, Runway Centerline & Runway End lights. TWY Marking: Taxi Side Stripe, Taxiway Centre line & Runway Holding positions, Intermediate holding position (except B3) and Enhanced Taxiway centreline markings. Lights: Taxiway edge Lights are available only at A, A1, A2, A4, A5, A6, A7, A8, A9, A10, A11, K, B3, B4, B6. Taxiway edge Lights are available at intersections of taxiways B, P, Q, G & H. Runway guard lights configuration A available at A1, A2, A4, A5, A6, A7, A8, A9, A10, A11, H1, H2, H3, H4, H5, H6, H7, H8, H9, H10.. RETILS's Yellow color, providing a 3-2-1 countdown Pattern are provided for RETs A5, A6, A7, A8, A9, H4, H5, H6, H7. IHP lights are provided at all IHP locations. Stop bars at IHP are provided at cross field taxiway junctions of TWY A, B, G, H, P, Q (at either sides of cross field taxiways). Taxiway centreline lights available on TWY A between A1 & B4, A1, A2, A4, B4, B, B1, B2, D, F, P, P1, M, N, N1, N2, Q, R, S, G, G1, G4, H, H1, H2, H3, H4, H5, H6, H7, H8, H9, H10. Taxiway centre line lights available on RET A6 & A7 till RWY holding position marking. Taxiway centre line lights available for RET A5 & A8 from RWY 09L/27R till RWY holding position. No Entry bar lights are provided for RET A5, A6, H4, H5 & H6, H7. No Entry bar lights are provided at RET A7, RET A8 and RET A9.
3	Stop bars (if any)	Runway 09L/27R – At Runway Holding Positions A1, A2, A4, A11. Runway 09R/27L – At Runway Holding Positions H1, H2, H3, H8, H9, H10. Stop Bar Lights on TWY A10 (Runway holding position is without Lead-in Lights)
4	Remarks	Advanced Surface Movement Guidance and Control System (ASMGCS) provided.

VOBL AD 2.10 AERODROME OBSTACLES

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	131155.9N 0774219.1E	3201 FT	NO	Arrester On ATC Building Top
In circling area and at AD	BUILDING	131155.7N 0774219.4E	3150 FT	LGTD	ATC Building
In circling area and at AD	OTHER	131156.0N 0774219.2E	3179 FT	LGTD	SMR 1 On ATC Building Top
In circling area and at AD	OTHER	131155.9N 0774219.2E	3174 FT	NO	Beacon On ATC Building Top
In circling area and at AD	OTHER	131124.2N 0774351.7E	2981 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131229.6N 0774126.5E	2998 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131229.6N 0774124.0E	2998 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131230.0N 0774123.4E	3007 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131228.7N 0774311.7E	2926 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131231.7N 0774116.8E	3022 FT	LGTD	WDI
In circling area and at AD	OTHER	131119.0N 0774135.1E	2981 FT	LGTD	GP Hut/Marked
In circling area and at AD	OTHER	131118.0N 0774325.6E	2974 FT	LGTD	GP Hut/Marked
In circling area and at AD	OTHER	131229.7N 0774120.0E	2999 FT	LGTD	GP Monitor Antenna (RWY 09L)/Marked
In circling area and at AD	OTHER	131229.6N 0774122.8E	3041 FT	LGTD	GP Antenna (RWY 09L)/Marked
In circling area and at AD	OTHER	131229.6N 0774122.8E	3045 FT	NO	Lightning Arrester on Glide Path Antenna
In circling area and at AD	OTHER	131229.5N 0774123.0E	3008 FT	NO	Antenna On Glide Path Hut
In circling area and at AD	OTHER	131229.4N 0774308.8E	2943 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131229.4N 0774309.1E	2937 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131228.7N 0774309.2E	2928 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131228.7N 0774309.7E	2927 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131228.6N 0774311.9E	2935 FT	NO	Antenna on GP Hut
In circling area and at AD	OTHER	131228.6N 0774312.1E	2972 FT	NO	Lightning Arrester on GP
In circling area and at AD	OTHER	131228.6N 0774312.1E	2968 FT	LGTD	GP Antenna (RWY 27R)/Marked

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	131228.6N 0774314.8E	2939 FT	LGTD	GP Monitor Antenna (RWY 27R)
In circling area and at AD	OTHER	131228.6N 0774312.0E	2930 FT	LGTD	GP Hut Top/Marked
In circling area and at AD	OTHER	131213.5N 0774205.2E	3082 FT	LGTD	Flood Light Mast
In circling area and at AD	OTHER	131126.9N 0774131.8E	2987 FT	LGTD	WDI
In circling area and at AD	OTHER	131119.1N 0774135.6E	2976 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131119.0N 0774136.6E	3004 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131119.1N 0774136.6E	2976 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131119.5N 0774212.0E	2971 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131119.5N 0774213.0E	2971 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131119.2N 0774247.6E	2967 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131119.2N 0774248.6E	2967 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131118.2N 0774324.1E	2969 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131118.1N 0774325.1E	2996 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131118.2N 0774325.1E	2969 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131125.8N 0774328.9E	2979 FT	LGTD	WDI
In circling area and at AD	OTHER	131229.5N 0774124.0E	2998 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131229.5N 0774125.0E	2997 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131228.7N 0774312.1E	2940 FT	LGTD	GP DME (RWY 27R)
In circling area and at AD	OTHER	131228.6N 0774310.2E	2927 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131228.6N 0774309.2E	2927 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131229.7N 0774124.1E	3024 FT	NO	Meteorological Equipment
In circling area and at AD	OTHER	131229.2N 0774158.8E	2982 FT	NO	Antenna
In circling area and at AD	OTHER	131228.8N 0774308.6E	2952 FT	NO	Meteorological Equipment

VOBL AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

1	Name of the associated meteorological office	Meteorological Centre, Bengaluru

2	Hours of service and, where applicable, the designation of the responsible meteorological office outside these hours	H24
3	Office responsible for preparation of TAFs and periods of validity and interval of issuance of the forecasts	Meteorological office at airport(IMD) 09 & 30 HRS
4	Availability of the trend forecast for the aerodrome and interval of issuance	Trend Trend Forecast appended to METAR/Met report
5	Information on how briefing and/or consultation is provided	H-24
6	Types of flight documentation supplied and language(s) used in flight documentation	Tabular and Chart Forms. English
7	Charts and other information displayed or available for briefing or consultation	Upper winds: FL050, FL100, FL140, FL180, FL240, FL300, FL340, FL390, FL450.
8	Supplementary equipment available for providing information on meteorological conditions, e.g. weather radar and receiver for satellite images;	Telefax. PWD 22 RVR instrument at RWY 09L and 27R with digital current weather instrument system (DCWIS). 09L MID and 27R MID commissioned and operational.
9	The air traffic services unit(s) provided with meteorological information	Bengaluru ATC and ACS
10	Additional information, e.g. concerning any limitation of service.	(i) Automatic Weather Station available for all Runway 09L/27R & 09R/27L. Integrated panel available at Met Briefing Room and Control Tower. Instrumental RVR reading available for all Runway.

VOBL AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

Designations	TRUE Bearings	Dimensions of RWY (M)	Strength of pavement (PCR) and associated data) and surface of runway and associated stopways	Geographical coordinates for threshold and runway end
1	2	3	4	5
09L	90.51 DEG	4000 x 45 M	888/F/B/W/T Asphalt	THR: 131225.79N 0774109.86E RWY END: 131224.65N 0774322.69E
27R	270.51 DEG	4000 x 45 M	888/F/B/W/T Asphalt	THR: 131224.65N 0774322.69E RWY END: 131225.79N 0774109.86E
09R	90.50 DEG	4000 x 45 M		THR: 131123.04N 0774123.92E RWY END: 131121.89N 0774336.74E
27L	270.50 DEG	4000 x 45 M		THR: 131121.89N 0774336.74E RWY END: 131123.04N 0774123.92E

THR elevation and highest elevation of TDZ of precision APP RWY	Slope of runway and associated stopway	Dimensions of stopway (M)	Dimensions of clearway (M)	Dimensions of strips (M)
6	7	8	9	10
THR: 3003.0FT TDZ: 3003.0FT	0.63%	NIL	NIL	4120 x 280 M
THR: 2919.0FT TDZ: 2937.0FT	0.63%	NIL	NIL	4120 x 280 M
THR: 2973.0FT TDZ: 2973.0FT	0.07%	NIL	NIL	4120 x 280 M
THR: 2964.0FT TDZ: 2965.0FT	0.07%	NIL	NIL	4120 x 280 M

Dimensions of runway end safety areas	Location and description of arresting system (if any)	Existence of an obstacle-free zone	Remarks.
11	12	13	14
240M x 90M	NA	AVBL	1. The surface of beginning 150m & 180m for runway 09L & 27R respectively are Concrete and PCR 1143/R/B/W/T 2. Gravity Model EGM-08. 3. PCR 888/F/B/W/T (FROM 150M TO 3820M)
240M x 90M	NA	AVBL	1. The surface of beginning 150m & 180m for runway 09L & 27R respectively are Concrete and PCR 1143/R/B/W/T 2. Gravity Mode EGM-08. 3. PCR 888/F/B/W/T (FROM 180M TO 3820M)
240M x 90M		AVBL	1.Gravity Mode EGM-08 2. The surface of 150 M of runway beginning at both ends is Concrete and PCR 1143/R/B/W/T 3. RWY PCR 1069/F/C/W/T (FM 150 m to 3850 m), ASPHALT
240M x 90M		AVBL	1. Gravity Model EGM-08 2. The surface of 150 M of runway beginning at both ends is Concrete and PCR 1143/R/B/W/T 3. RWY PCR 1069/F/C/W/T (FM 150 m to 3850 m) ASPHALT

VOBL AD 2.13 DECLARED DISTANCES

RWY Designator	Take-off run available TORA (M)	Take-off distance available TODA (M)	Accelerate distance available ASDA (M)	Landing distance available LDA (M)	Remarks (including runway entry or start point where alternative reduced declared distances have been declared)
1	2	3	4	5	6
09L	4000	4000	4000	4000	
27R	4000	4000	4000	4000	
09R	4000	4000	4000	4000	
27L	4000	4000	4000	4000	

VOBL AD 2.14 APPROACH AND RUNWAY LIGHTING

Runway Designator	Type, length and intensity of approach lighting system	Runway threshold lights, colour and wing bars	Type of visual slope indicator system	Length of runway touchdown zone lights
1	2	3	4	5
09L	CAT I 900 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (64.97FT)	
27R	CAT I 900 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (65.35FT)	
09R	CAT III 900 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (65.89FT)	900 M
27L	CAT III 900 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (65.89FT)	900 M

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
4000 M 15 M LIH	3400M 60M White 600M 60M Yellow LIH	Red		a) Runway centreline lights White from the threshold to the point 900m from the runway end, alternate red and white from 900m to 300m from the runway end and red from 300m to the runway end. b) PAPI are connected with interleaving circuits
4000 M 15 M LIH	3400M 60M White 600M 60M Yellow LIH	Red		a) Runway centreline lights White from the threshold to the point 900m from the runway end, alternate red and white from 900m to 300m from the runway end and red from 300m to the runway end. b) PAPI are connected with interleaving circuits

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
4000 M 15 M LIH	3400M 60M White 600M 60M Yellow LIH	Red		a) Runway centreline lights White from the threshold to the point 900m from the runway end, alternate red and white from 900m to 300m from the runway end and red from 300 m to the runway end. b) PAPI are connected with interleaving circuits c) Sequenced flashing lights supplementing ALS available
4000 M 15 M LIH	3400M 60M White 600M 60M Yellow LIH	Red		a) Runway centreline lights White from the threshold to the point 900m from the runway end, alternate red and white from 900m to 300m from the runway end and red from 300m to the runway end. b) PAPI are connected with interleaving circuits c) Sequenced flashing lights supplementing ALS available

VOBL AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

1	Location, characteristics and hours of operation of aerodrome beacon/identification beacon (if any)	ABN	Top of the control tower - Green and white flashing lights. 24 FPM/12 RPM
		IBN	Not AVBL
2	Location and lighting (if any) of anemometer/ landing direction indicator;	LDI	Runway 09L/27R – (i) Landing T provided south of Taxiway A and east of Taxiway-B4 and illuminated. (ii) WDI provided at both ends of Runway north of Threshold 09L/27R & East of Taxiway-B4 south of Taxiway A. Runway 09R/27L – (i) WDI provided at both ends of Runway north of runway threshold 09R/27L. (ii) All Wind Direction Indicator are illuminated.
		Anemometer	Anemometer located on top of administrative block
3	Taxiway edge and taxiway centre line lights;	Edge	Taxiway edge Lights are available only at A, A1, A2, A4, A5, A6, A7, A8, A9, A10, A11, K, B3, B4 and B6. Taxiway edge Lights at curves are available at intersections of taxiways B, P, Q, G & H.
		Centre Line	Taxiway centreline lights are available for TWY A, A1,A2,A4,A11,K,B4,B,B1,B2,D,F,P,P1,M,N,N1,N2, Q,R,S, G,G1,G4,H,H1,H2,H3,H4,H5,H6,H7,H8,H9,H10. Taxiway centre line lights available on RET A5,A6,A7,A8 from runway till RWY holding position marking.

4	Secondary power supply including switch-over time;	Runway 09L/27R - All ground lights are connected to UPS except Taxiway edge lights and taxiway centreline lights. Two 425KVA DG sets available one each at 09L and 27R stations for ground lighting with switch over time of 15 sec. For apron lighting 1500KVA DG sets 05 nOS.. are available. Runway 09R/27L - 02 numbers of 1250KVA prime power rated DG sets (with 100% redundancy/standby) at each of the AGL substations (09R CCR & 27L CCR) available for Airfield ground lighting of Runway 09R/ 27L and associated taxiways. Complete AGL lighting of Runway 09R/27L and crossfield taxiways including taxiway edge & airfield guidance signs are connected in on-line UPS with switch over time less than one second..
5	Remarks	Taxiway centre line lights are available on Apron for Taxilane L1, L2, L3, C, D, B1, M, N & N1. Trifurcation lights are available on Apron at Taxilane K at eastern side, Taxilane B4 and B2 on either side with selective control. Taxilane L1 & L3 lights will be switched ON for Code-C & below and Taxilane L2 lights will be switched ON for Code-D and above. Straight portion of taxilane centre line lights for Taxilane L1, L2 & L3 will remain ON.

VOBL AD 2.16 HELICOPTER LANDING AREA

1	Geographical coordinates of the geometric centre of touchdown and lift-off (TLOF) or of each threshold of final approach and take-off (FATO) area	Not Established
2	TLOF and/or FATO area elevation:	Not Established
3	TLOF and FATO area dimensions to the nearest metre or foot, surface type, bearing strength and marking;	Not Established
4	True bearings of FATO;	Not Established
5	Declared distances available	Not Established
6	Approach and FATO lighting;	Not Established
7	Remarks	Not Established

VOBL AD 2.17 AIR TRAFFIC SERVICE AIRSPACE

1	Airspace designation, geographical coordinates and lateral limits	CTR: Area bounded by 130100N 0771410E then along clockwise arc of 30NM centered at 131224N 0774356E to 130000N 0781015E to 130000N 0780000E to 130100N 0773706E to point of origin (except the restricted airspace of VOR 185A, VOR 185B and VOR185C)
2	Vertical limits	FL 95
3	Airspace classification	D
4	Call sign and language(s) of the air traffic services unit providing service;	Bengaluru Tower / Bengaluru Radar, English
5	Transition altitude	7000 FT
6	Hours of applicability	H24
7	Remarks	Airspace from 7000 FT AMSL and above is Classified as Class ‘C’.

VOBL AD 2.18 AIR TRAFFIC SERVICES COMMUNICATION FACILITIES

Service Designation	Call sign	Channel(s)	SATVOICE Number(s), if available
1	2	3	4
OTHER	Bengaluru Delivery	121.825 MHZ	
OTHER	Bengaluru Delivery	121.900 MHZ	
APP	Bengaluru Radar	121.250 MHZ	
APP	Bengaluru Radar	125.275 MHZ	
APP	Bengaluru Radar	127.750 MHZ	
TWR	Bengaluru Tower	118.475 MHZ	
TWR	Bengaluru Tower	119.050 MHZ	
TWR	Bengaluru Tower	123.675 MHZ	
TWR	Bengaluru Tower	124.350 MHZ	
ATIS	Bengaluru information	128.675 MHZ	
ACC	Bengaluru Radar	119.450 MHZ	
ACC	Bengaluru Radar	125.750 MHZ	
ACC	Bengaluru Radar	126.325 MHZ	
ALRS	----	121.500 MHZ	
SMC	Bengaluru Ground	121.650 MHZ	
SMC	Bengaluru Ground	121.775 MHZ	

Logon address, as appropriate	Hours of operation	Remarks
5	6	7
	0000 - 1730 UTC	CLD-1, MAIN FREQUENCY Beyond this period clearance Delivery will on Freq 121.775 MHz
	0000-1730 UTC	CLD-2, BACKUP FREQUENCY Beyond this period clearance Delivery will on Freq 121.775 MHz
	H24	APPROACH ARRIVAL, MAIN FREQUENCY
	H24	APPROACH DEPARTURE APPROACH ARRIVAL BACKUP FREQUENCY
	H24	APPROACH DEPARTURE, MAIN FREQUENCY
	H24	TWR – 2, BACKUP FREQUENCY
	H24	TWR – 2, MAIN FREQUENCY
	H24	TWR – 1, BACKUP FREQUENCY
	H24	TWR – 1, MAIN FREQUENCY
	H24	ATIS VOICE SERVICE IS AVBL ON TEL NO. 080-22019936.
	H24	LOWER BENGALURU (SECTOR-WEST) MAIN FREQUENCY
	H24	LOWER BENGALURU BACKUP FREQUENCY
	H24	LOWER BENGALURU (SECTOR-EAST) MAIN FREQUENCY
	H24	EMERGENCY FREQUENCY
	H24	SMC – 1, MAIN FREQUENCY
	H24	SMC – 2, MAIN FREQUENCY

VOBL AD 2.19 RADIO NAVIGATION AND LANDING AIDS

Type of aids, magnetic variation and type of supported operation for ILS/ MLS, basic GNSS, SBAS and GBAS, and for VOR/ILS/MLS station used for technical lineup of the aid	Identification	Frequency(ies), Channel number(s), Service provider, and reference path identifier(s) (RPI), as appropriate	Hours of operation, as appropriate;
1	2	3	4
LOC 27R CAT I	IDEV	108.300 MHz	As ATS
LOC 09L CAT I	IBAN	109.300 MHz	As ATS
LOC 27L CAT IIIB	IDVN	111.500 MHz	H24
LOC 09R CAT IIIB	IBLN	109.900 MHz	H24
GP 27R		334.100 MHz	As ATS
GP 09L		332.000 MHz	As ATS
GP 27L		332.900 MHz	H24
GP 09R		338.800 MHz	H24
DME ILS 09L	IBAN	CH30X	As ATS
DME ILS 27R	IDEV	CH20X	As ATS
DME ILS 27L	IDVN	CH52X	H24
DME ILS 09R	IBLN	CH36X	H24
DVOR/DME	BIA	116.800 MHz CH115X	H24

Geographical coordinates of the position of the transmitting antenna	Elevation of transmitting antenna of DME/ elevation of GBAS reference point	Service volume radius from the GBAS reference point	Remarks
5	6	7	8
131225.9N 0774059.4E			
131224.6N 0774333.1E			
131123.2N 0774112.3E			
131121.7N 0774351.3E			
131228.7N 0774312.1E			1. Coverage of GP is Restricted to plus or minus 8 DEG at 1500 FT AGL at 8NM from THR and plus or minus 8 DEG at 2000FT AGL at 10NM from THR 2. Glide Angle 3.0 DEG

131229.7N 0774122.8E			1. Coverage of GP Restricted to plus or minus 7 Deg at 10NM 2. Glide Angle 3.0 DEG
131118.0N 0774325.8E			
131119.0N 0774134.8E			
131229.8N 0774122.7E	3021 FT		Elevation in EGM-08
131228.8N 0774312.1E	2943 FT		Elevation in EGM-08
131117.8N 0774325.8E	2977 FT		Elevation in EGM-08
131118.8N 0774134.9E	2984 FT		Elevation in EGM-08
131232.9N 0774140.9E	3025 FT		Elevation in EGM-08

VOBL AD 2.20 LOCAL AERODROME REGULATIONS

I.Yelahanka (VOYK) / Military Airbase exists at close proximity of South of Kempegowda International Airport, Bengaluru where Military Flying activities are conducted.

Orientation	Location	Helicopter Flying Activities
090/270 DEG MAG	3.2NM south of extended centre line of VOBL Runway 09R/27L	Helicopter Traffic Circuit upto 1.5NM North of VOYK RWY

Remarks:

1. Pilots landing at VOBL to exercise caution for VOYK having similar Runway orientation and is 4.3 NM south of VOBL Runway 09L/27R and 3.2NM south of VOBL Runway 09R/27L.
2. Visual operations by Military helicopters at VOYK, Height not exceeding 700ft AGL and circuit not exceeding 1.5 NM North of VOYK Runway.
3. No left turn after missed approach from 27L, No right turn after missed approach from 09R and Departures to strictly follow promulgated SIDS due to proximity of VOYK.
4. All ACFT OPR on RWY 09R/27L VOBL to EXER CTN due
   i. Practice ILS approaches by military TRG FLT AT VOYK RWY27, 3.2NM South of extended centreline of RWY 09R/27L VOBL.
   ii. Fixed wing ACFT OPS upto 6000FT in VOYK VOR 185 A/B/C).

VOBL AD 2.21 NOISE ABATEMENT PROCEDURES

Consistent with safety of aircraft operations and in consideration of high intensity runway operations, pilots should minimize the use of reverse thrust after landing to reduce disturbance in areas adjacent to the aerodrome.

VOBL AD 2.22 FLIGHT PROCEDURES

I.TRANSPONDER OPERATING PROCEDURES ON GROUND

1.Introduction:

Advanced Surface Movement Guidance and Control System (A-SMGCS) using Mode-S Multi-lateration has been commissioned at Bangalore, Chennai, Delhi, Hyderabad, Kolkata and Mumbai International Airports.

The Aircraft Transponder Operating Procedures, particularly in the movement area of the airport(s), where A-SMGCS has been commissioned, is as given below:

2.DEPARTURE

iAt the Gate/Stand:

Select STBY. Enter the discrete SSR code received from Clearance Delivery/Surface Movement Control. Enter the three letter ICAO designator followed by the flight identification number (e.g. AIC748) through the FMS or the Transponder control panel, depending on the avionics.

ii.On requesting Pushback/Taxi (whichever is earlier):

Select Transponder or equivalent and AUTO if available. This action will enable the aircraft ID, used as the Call sign by ATC, to be displayed on the surveillance display of ATC. ATC can verify the data and use it for necessary identification.

iii. When Lining up:

Select TCAS. Select TCAS only after receiving the clearance to line up, to ensure that the performance of systems based on SSR frequencies (including airborne TCAS units, SSR and A-SMGCS) is not compromised.

3.ARRIVAL

1.When on the Runway:

Keep TCAS selected

2.After vacating the Runway:

Select Transponder or equivalent and AUTO if available. There is a need that the Transponder remains able to exchange data with the A-SMGCS system. However to ensure that the performance of systems based on SSR frequencies (including airborne TCAS Unit, SSR and A-SMGCS) is not compromised, TCAS shall be deselected when vacating the Runway.

3.Parked on Stand:

Select STBY. The Transponder will not reply to interrogation. The discrete SSR Code given to that particular flight can now be recycled for other flights.

Note: When on ground the aircraft must squawk Mode C, in order to provide the altitude information to the surveillance system, and thereby prevent:

i) clutter on Terminal Approach Radar Display (and)
ii) false automatic detection of departure for aircraft still on ground.

II. LOW VISIBILITY PROCEDURE

1. DEFINITIONS & ABBREVIATIONS

1.1 Low Visibility operations (LVO). Approach operations in RVRs less than 550 m and/ or with a DH less than 60 m (200 ft) or take-off operations in RVRs less than 400 m.

1.2 Low Visibility Procedures (LVP): Specific procedures applied at an aerodrome for the purpose of ensuring safe operations during Categories II and III approaches and/or low visibility take-offs.

1.3 Low Visibility Take-Off (LVTO): A term used in relation to flight operations referring to a take-off on a runway where the RVR is less than 400m.

1.4 Safeguarding Procedures: Safeguarding Procedures (SP) are instructions for relevant airport departments and airside operators to prepare ground services and facilities for low visibility operations, in order that when LVP are implemented all SP are complete and airport is configured for Low Visibility Procedures and Low Visibility Take-offs.

1.5 Segregated parallel operations: Simultaneous operations on parallel or near-parallel instrument runways in which one runway is used exclusively for approaches and the other runway is used exclusively for departures.

a) A segregated mode of operation with Runway 27L is used for landing and Runway 27R is used for departures is termed as 27 Segregated Mode.

b) A segregated mode of operation with Runway 09L is used for landing and Runway 09R is used for departures is termed as 09 Segregated Mode.

1.6 09 Dependent mode: Runway 09 is used in dependent segregated mode wherein RWY 09R is used for arrivals and RWY 09L is used for departures

2. INTRODUCTION:

2.1 MODES OF OPERATIONS AT KIAB

2.1.1 09 Segregated Mode: Runway 09 is in use in segregated mode wherein RWY 09L is used for arrivals and RWY 09R is used for departures.

2.1.2 27 Segregated Mode: Runway 27 is in use in segregated mode, wherein RWY 27L is used for arrivals and RWY 27R is used for departures.

2.1.3 09 Dependent mode: Runway 09 is used in dependent segregated mode wherein RWY 09R is used for arrivals and RWY 09L is used for departures.

2.1.4 09R Mode or 27L Mode: Single south runway (09R or 27L) is in use for arrivals and departures . Accordingly it is termed 09R mode or 27L mode.

2.1.5 09L Mode or 27R Mode: Single north runway (09L or 27 R) is in use for arrivals and departures. Accordingly it is termed 09L mode or 27R mode.

2.1.6 During LVP**,** mode of operations shall be 27 Segregated Mode or 09 Dependent Mode or Single RWY 09R/27L mode.

2.1.7 Whenever a pilot is requesting LVO, based on low cloud ceiling, south RWY 09R or 27L shall be used for arrivals and LVP is to be imposed.

2.1.8 RWY 09R/27L is CAT I/CAT II/CAT III compliant and RWY 09L/27R is CAT I compliant.

2.2 GENERAL

2.2.1 Safeguarding Procedures(SP)/Low visibility Procedure will be applicable for the entire Aerodrome.

2.2.2 In case of abandoned take off, mandatory follow-me service shall be provided on north runway for RET A5/A6/A7/ A8/A9 and TWY A10 till joining TWY A whenever RVR is 350m or less. (RET A5/A6/A7/A8 taxiway centre line lights are provided from runway till runway holding postion marking only. TWY-A10 and RET A9 are not provided with taxiway centre line lights).

2.2.3 The FOG PDS (Post fog departure sequence) issued by AOCC shall be followed. International flights will be kept out of this sequencing due few in numbers and longhaul flights shall be accommodated as and when start-up is requested as per SOBT.

2.2.4 The following equipments shall be serviceable to the required standard to support CAT II/CAT III operations: -

a) ILS localizer Transmitters (Both Main and Stand by).
b) ILS glide path Transmitter (Both Main and Stand by).
c) ILS DME or Outer Marker.
d) ILS Localiser Far Field Monitors (For CAT III only).
e) Airfield ground lighting system (AGL)
f) Instrumented RVR system: (TDZ, MID & END for CAT III; TDZ & MID for CAT II).
g) Standby power supply for ILS.
h) UPS/ Standby DG Set power for aeronautical ground lighting system.
i) Surface Movement Radar SMR for CAT II/CAT III operations/ LVTO (for RVR below 350M).

(DGCA CAR Section 4 Series B Part 1, Para 9.8.7. — Surface movement radar for the manoeuvring area shall be provided at an aerodrome intended for use in runway visual range conditions less than a value of 350 m.)

When any equipment and facilities listed in Para 2.2.4 above become unserviceable during periods of LVP, the concerned ATC unit shall advise all aircraft and accordingly suspend CAT II/ CAT III/ LVTO operations. Information to this effect shall be included in ATIS broadcast.
The pilot shall ensure that he is suitably qualified and certified to carry out the required category of ILS approach.

2.3 ILS SENSITIVE AND CRITICAL AREAS

2.3.1 The critical and sensitive areas of ILS have been shown in eAIP AD 2.24. Protection of these areas during CAT-II/ CAT-III operations will be ensured as per Standard Operating Procedures.

Note: Signage indicating the limits of localizer and glide paths sensitive areas are provided.

2.3.2 Diagram indicating the Critical and Sensitive area of ILS for all runways shall be available with Airside Operations , control tower and WSO.

2.3.3 ILS Critical and Sensitive areas are depicted and it is indicated that TWY-H1 and H2 are not to be used when ILS RWY 09R is in operation and TWY-H9 and H10 are not to be used when ILS RWY 27L is in operation.

3. IMPLEMENTATION OF SAFEGUARDING PROCEDURES (SP)

3.1 Criteria for Safeguarding Procedures: Safeguarding Procedures shall be initiated when:
a) The RVR is less than 1200 m and/or visibility is forecast to deteriorate to 800 m or less; and/or
b) The cloud ceiling is 400 ft and is forecast to fall to 200 ft or less.

3.2 Process of initiation of Safeguarding Procedures:

3.2.1 When meteorological conditions meet the criteria for initiation of SP as stipulated in Para 3.1, the Tower Supervisory Officer will inform the Airside Ops Manager/AOCC, ARFF, the SSO and the WSO.

3.2.2 When all the concerned agencies have completed their necessary actions, they shall report to Tower Supervisory Officer that their Safeguarding Procedures (SP) is completed and the airport is safeguarded for LVP operations. TSO will, inturn, inform WSO.

4. CRITERIA FOR IMPLEMENTATION OF LOW VISIBILITY PROCEDURES:

4.1 Low Visibility Procedures shall be implemented when

a) Either TDZ, MID or END RVR is less than 800 m; and/or Cloud ceiling is less than 200 ft; and
b) Safeguarding Procedures (SP) have been completed and the airport is safeguarded.
Note 1: Whenever any Pilot is requesting for LVP based on low cloud ceiling, LVP will be imposed.
Note 2: Though LVP is implemented when RVR is less than 800m, ILS CAT I operation will continue till TDZ RVR is not less than 550m.
5. APRON MEASURES

5.1 Lane change (L1-L3/L3-L1) is not permitted for taxing aircraft when RVR reduces below 200 m/ at the discretion of Airside Operations Manager.

5.2 The decision to stop lane changes above RVR 200m (due to poor visibility in the apron) will be at the sole discretion of Airside Operations Manager and same will be conveyed to WSO/TWR SUP immediately.

5.3 Only one aircraft towing is permitted when LVP is implemented. Lane change for towing aircraft will be under the guidance of FM service only.

5.4 Vehicle control measures on Apron shall be implemented when visibility is reported at or below 200 meters on the Apron as observed by the Airside Operations Manager/MET.

5.5 Mandatory Follow-Me service shall be provided for all arriving aircraft to the stand not fitted with ASMGL when RVR is below 350m and, as and when requested by Pilot/ATC.

6. TERMINATION OF LOW VISIBILITY PROCEDURES AND SAFEGUARDING PROCEDURE

6.1 WSO may terminate LVP when:
a) Meteorological conditions improve and TDZ, MID & END RVR are 800 m or more and the cloud ceiling is 200 ft or higher,
and trend is for improvement.
b) Facilities and equipment (listed in Para 2.2.4) necessary for CAT II/CAT III operations are degraded and/ or the prevailing conditions are considered unsafe for such operations.

6.2 WSO should consult Meteorological Office for forecast before cancelling LVP.

6.3 The WSO will intimate the following regarding the termination of LVP operations:
a) Tower Supervisory Officer
b) Approach/Radar Controller
c) Shift Supervisory Officer (SSO)

6.4 Tower Supervisory Officer will in turn inform AOCC/ Airside Ops Manager, who will advise all the previously notified personnel to resume normal operations.

6.5 On cancelling LVP, TSO shall include it in the subsequent two DATIS broadcasts that "LOW VISIBILITY PROCEDURES ARE
CANCELLED". Tower Controller/TSO will inform all the concerned agencies.

6.6 TSO may terminate Safeguarding procedures when:
a) Meteorological conditions improve and TDZ, MID & END RVR are 1200 m or more and
b) The cloud ceiling is 400 ft or higher, and trend is for improvement.

6.7 If SP are completed, and LVP is not subsequently implemented and meteorological conditions improve and the RVR is more than
1200 m and the cloud ceiling is 400 ft or higher and both are forecast to remain above the required SP criteria, TSO may cancel SP.

7. PUSH BACK RESTRICTION

7.1 While operating Single Runway Mode at Low visibility (below RVR 550 M), the pushbacks may be restricted as follows:
7.1.1 Maximum three pushback simultaneously on each Apron (Apron 1,2,3,4).
7.1.2 But limited to maximum four pushback simultaneously on Apron 1& Apron 2 combined and Apron -3 & Apron -4.
Note: When 27 Segregated mode / 09 Dependent mode is in use, no pushback restrictions are required till lane change restrictions are imposed by AOM or RVR falls below 125m.
7.2 These restrictions are aimed at reducing holding point congestion during CAT II/III operations as the spacing between the departing flights is more to clear the ILS critical and sensitive areas.
7.3 In case ASMGCS is not available, the pushback restrictions will be imposed by Tower Supervisor in coordination with WSO dynamically.

8. EQUIPMENT AND SERVICES

8.1 Runway Visual Range (RVR)
(a) Four transmissometers are available for South Runway (09R/27L) designated as 27L TDZ, 27L MID, 09R MID & 09R TDZ,
09R MID RVR reading will be used as END RVR for RWY 27L and 27L MID RVR reading will be used as END RVR for
RWY 09R.
(b) Four transmissometers are available for North Runway (09L/27R) designated as 27R TDZ, 27R MID, 09L MID & 09L TDZ.
09L MID RVR reading will be used as END RVR for RWY 27R and 27R MID RVR reading will be used as END RVR for
RWY 09L.
(c) RVR is reported in the following scales:
(i) From 2000m to 1200m, increments of 100m
(ii) From 1200m to 800m, increments of 50m
(iii) From 800m to 50m, increments of 25m
(d) Equipment serviceability for CAT II/CAT III operations:
(i) For CAT II operations TDZ and MID RVR shall be available
(ii) For CAT III operations TDZ, MID and END RVR shall be available.

8.2 Reporting of RVR Values

8.2.1 Touch-down zone RVR needs to be reported for CAT I operations; touch-down and mid zone RVR for CAT II operations;
touch-down, mid and roll-out zone RVR for CAT III operations. In all cases, touch-down zone RVR will always be controlling
RVR. However, if any other RVR is reported and is relevant (operator shall not define relevant depending on Runway length/aircraft stopping distance unless approved by FSD, DGCA) it also becomes controlling. The mid-zone and roll-out zone can be
lower than the touch-down zone provided conditions enumerated in Note 1 below are satisfied. The following RVR values mentioned in Table 3 should be used for reference.

Type of Operation	**RVR**		
	Touch-down Zone	**Mid Zone**	**Roll-out Zone**
CAT I	550M	125M	125M
CAT II	300M	125M	125M
CAT III	175M	125M	125M
CAT III (with roll-out guidance)	75/50M	75/50M	75/50M

Note 1: The use of minimum RVRs in the table above is subject to:

- Operator authorization.
- Aeroplane authorization.
- Flight crew training and qualification; and
- Aerodrome facilities.

Note 2: The use of minimum RVR of 75m or 50m depends on value approved for operators and aeroplanes with roll-out guidance system.

Note 3: The values in bold font are required for the type of operation.

9. ACTIONS BY SHIFT SUPERVISORY OFFICER (SSO)

9.1 On receipt of "Outlook for LVP" from TSO, the Shift Supervisory Officer (SSO) will inform the Duty Officer, NAV AIDS and have the ILS equipment and its status indicators in ATC units checked up. He will inform WSO of any unserviceability in the equipment which is likely to affect ILS CAT II/CAT III operation.

9.2 On receipt of "Advisory Message" from WSO that LVP is to be made effective, SSO will maintain continuous watch on the performance of ILS equipment and will inform WSO of any un-serviceability which may affect ILS CAT II/CAT III operation.

10. ACTIONS BY DUTY OFFICER (METEOROLOGICAL OFFICE)

10.1 Duty Meteorological Officer would issue an 'Outlook for Low Visibility Procedures' to the Watch Supervisory Officer (WSO) of air traffic services whenever he expects that the RVR and/or cloud ceiling will fall below 800 m and/or 200ft or less respectively.

10.2 Whenever Duty Meteorological Officer visualizes that RVR is likely to fall below 800m and/or cloud ceiling to 200ft or less within next 2 hours, he will issue an 'Advisory Message' to WSO to this effect.

10.3 When the RVR and/or cloud ceiling are 800m and/or 200ft respectively and the trend is towards improvement in these elements of weather conditions the Duty Met Officer may, when requested by WSO, advise him about such improving weather conditions
for the purpose of termination of LVP.

10.4 The Duty Met Officer would ensure that the RVR displays in ATC units in the Control Tower and Approach Control are serviceable. He would also ensure that RVR/visibility recorders of Touch-down zone, Mid-point and End positions are serviceable.

10.5 Manual RVR in case of unserviceability of RVR equipment as and when required.

III) RADIO COMMUNICATION FAILURE PROCEDURES

1. INTRODUCTION:

Radio communication failure procedures are described in Para 15.3 of ICAO PANSATM DOC 4444. Based on these provisions, following radio communication failure procedures are established, to standardize the actions to be taken by the pilot of arriving and departing aircraft at Kempegowda International airport.

2. GENERAL:

2.1. All Transponder equipped aircraft experiencing Radio Communication failure shall set transponder to Mode A/C code 7600 as soon as practicable.

(Note: *This Requirement of Setting transponder to Mode A/C code 7600 in no way imposes any restriction on the pilot's decision to set transponder to Mode A/C code 7500 or 7700, whenever require*d).

2.2 For arriving aircraft, when the landing runway has been advised by ATC, that runway shall be considered the assigned runway. If no runway has been advised by ATC, Runway 09L shall be considered the assigned landing runway for arrival. During notified periods of maintenance or closure of Runway 09L, Runway 09R shall be considered the assigned landing runway.

2.3 In strong tail wind conditions for assigned Runway, aircraft experiencing RCF will carry out a missed approach on the assigned runway and after following the complete missed approach for such approach, aircraft will carry out published ILS approach procedure for the opposite Runway. (eg:-Runway 27R in case of assigned Runway 09L and Runway 27L in case of assigned Runway 09R and vice versa).

2.4 In case of departures from Bengaluru returning on account of RCF, the departure runway of such aircraft shall be considered as assigned runway for landing.

2.5 If Radio communication failure occurs after the final/ interception turn, the aircraft may continue its approach to land on such Runway.

2.6 Runway lights, PAPI, ILS, and Approach lights in 'SWITCHED ON' position shall indicate the availability of such Runway for aircraft experiencing RCF.

2.7 Irrespective of visibility/weather conditions, Runway and Approach lights in 'SWITCHED OFF' position shall indicate non-availability of such Runway for aircraft experiencing RCF. In such cases, alternate runway shall be made available for such aircraft. Runway 09R shall be alternate to Runway 09L and Runway 27L shall be alternate to Runway 27R.

2.8 In Kempegowda International Airport, When both runways are available for use, 27 mode means either RWY 27R or 27L will be used for Arrivals and 09 mode means either RWY 09L or 09R will be used for arrivals.

3. PROCEDURE FOR ARRIVALS:

3.1 Pilots of RNAV1 equipped Arrivals shall;

3.1.1 When Arrival clearance has been received from ATC follow the RCF procedure corresponding to that STAR as mentioned in para 3.1.3. & 3.1.4

3.1.2 When Arrival clearance has not been received from ATC follow the appropriate RNAV STAR corresponding to that route thereafter follow the respective RCF procedure as given in 3.1.3. &3.1.4

3.1.3 ARRIVAL ROUTE-SPECIFIC RCF PROCEDURES FOR RUNWAY 27L/ 27R:

At the end of the STAR, make a convenient turn to intercept localizer/final approach track of the published instrument approach procedure of the runway used for the arrivals, to carry out the approach procedure. At the end of the STAR, make a convenient turn to intercept localizer/final approach track of the published instrument approach procedure of the runway used for the arrivals, to carry out the approach procedure.

W81/N563 (GUNIM Transition)

a) when cleared via PEXEG7N/GUNIM7N maintain the last assigned level if,

i) at or below FL170, follow PEXEG 7N/GUNIM7N arrival with lateral and vertical restrictions.

ii) above FL170, proceed to ATPUK, descend in ATPUK hold to FL170 and follow PEXEG 7N/GUNIM 7N arrival with lateral and vertical restrictions.

b) Else maintain the last assigned level if,

i) at or below FL170, follow PEXEG 7F/GUNIM7F arrival with lateral and vertical restrictions.

ii) above FL170, proceed to ATPUK, descend in ATPUK hold to FL170 and follow PEXEG 7F/GUNIM 7F arrival with lateral and vertical restrictions.

Q9

Maintain the last assigned level if,

a) at or below FL150, follow RIKBU 7F arrival with lateral and vertical restrictions.

b) above FL150, proceed to RIGBA, descend in RIGBA hold to FL150 and follow RIKBU 7F arrival with lateral and vertical restrictions.

W78/W101E

a) When cleared via SUSIK 7L, Maintain the last assigned level if,

i) at or below FL150, follow SUSIK 7L arrival with lateral and vertical restrictions.

ii) above FL150, proceed to RIGBA, descend in RIGBA hold to FL150 and follow SUSIK 7L arrival with lateral and vertical restrictions.

b) Else Maintain the last assigned level if,

i) at or below FL150, follow SUSIK 7F arrival with lateral and vertical restrictions.

ii) above FL150, proceed to RIGBA, descend in RIGBA hold to FL150 and follow SUSIK 7F arrival with lateral and vertical restrictions.

W153/Q21

Maintain the last assigned level if,

a) at or below FL120, follow TELUV 7F arrival with lateral and vertical restrictions.

b) above FL120, proceed to TELUV, descend in RURPA hold to FL120 and follow TELUV 7F arrival with lateral and vertical restrictions.

W117/W142

Maintain the last assigned level if,

a) at or below FL120, follow XIVIL 7F arrival with lateral and vertical restrictions.

b) above FL120, proceed to DOXIT, descend in DOXIT hold to FL120 and follow XIVIL 7F arrival with lateral and vertical restrictions.

W42

Maintain the last assigned level if,

a) at or below FL150, follow ADKAL7F arrival with lateral and vertical restrictions.

b) above FL150, proceed to GURBI, descend in GURBI hold to FL150 and follow ADKAL7F arrival with lateral and vertical restrictions.

N563 (LEKAP Transition)

Maintain the last assigned level if,

a) at or below FL150, follow LEKAP7F arrival with lateral and vertical restrictions.

b) above FL150, proceed to GURBI, descend in GURBI hold to FL150 and follow LEKAP7F arrival with lateral and vertical restrictions.

W43

Maintain the last assigned level if,

a) at or below FL150, follow UGABA7F arrival with lateral and vertical restrictions.

b) above FL150, proceed to MUGPA, descend in MUGPA hold to FL150 and follow UGABA7Farrival with lateral and vertical restrictions.

3.1.4 ARRIVAL ROUTE–SPECIFIC RCF PROCEDURES FOR RUNWAY 09L/09R

At the end of the STAR, make a convenient turn to intercept localizer/final approach track of the published instrument approach procedure of the runway used for the arrivals, to carry out the approach procedure.

W81

Maintain the last assigned level if,

a. at or below FL150, follow PEXEG7E arrival with lateral and vertical restrictions.

b. above FL150, proceed to BL807, descend in BL807 hold to FL150 and follow PEXEG7E arrival with lateral and vertical restrictions.

N563 (GUNIM Transition)

Maintain the last assigned level if,

a. at or below FL150, follow GUNIM7E arrival with lateral and vertical restrictions.

b. above FL150, proceed to BL807, descend in BL807 hold to FL150 and follow PEXEG7E arrival with lateral and vertical restrictions.

O9

Maintain the last assigned level if,

a. at or below FL140, follow RIKBU7E arrival with lateral and vertical restrictions.

b. above FL140,proceed to BL805, descend in BL805 hold to FL140 and follow RIKBU7E arrival with lateral and vertical restrictions.

W78/W101E

If cleared for SUSIK 7J

Maintain the last assigned level if,

a). at or below FL120, follow SUSIK7J arrival with lateral and vertical restrictions.

b). Above FL120, proceed to RIGBA, descend in RIGBA hold to FL120 and follow SUSIK7J arrival with lateral and vertical restrictions.

Else,

Maintain the last assigned level if,

a) at or below FL140, follow SUSIK 7E arrival with lateral and vertical restrictions.

b) above FL140, proceed to BL805, descend in BL805 hold to FL140 and follow SUSIK 7E arrival with lateral and vertical restrictions.

W153

Maintain the last assigned level if,

a) at or below FL150, follow TELUV 7E arrival with lateral and vertical restrictions.

b) above FL150, proceed to RURPA, descend in RURPA hold to FL150 and follow TELUV 7E arrival with lateral and vertical restrictions.

W117/W142

Maintain the last assigned level if,

a) at or below FL150, follow XIVIL7E arrival with lateral and vertical restrictions.

b) above FL150, proceed to DOXIT, descend in DOXIT hold to FL150 and follow XIVIL7E arrival with lateral and vertical restrictions.

W42/N563(LEKAP Transition)

Maintain the last assigned level if,

a) at or below FL150, follow ADKAL 7E / LEKAP 7E arrival with lateral and vertical restrictions.

b) above FL150, proceed to GURBI, descend in GURBI hold to FL150 and follow ADKAL 7E / LEKAP 7E arrival with lateral and vertical restrictions.

W43

Maintain the last assigned level if,

a) at or below FL150, follow UGABA 7E arrival with lateral and vertical restrictions.

b) above FL150, proceed to MUGPA, descend in MUGPA hold to FL150 and follow UGABA 7E arrival with lateral and vertical restrictions.

3.1.5 ARRIVING AIRCRAFT BEING RADAR VECTORED

Aircraft being radar vectored for the approach, on experiencing RCF, shall maintain last assigned level and heading for 3 minutes after detecting RCF or selecting MODE A/C 7600, whichever is later (If below FL90, climb and reach FL90 before crossing 25NM outbound from BIA VOR) and then take a convenient turn to proceed direct to BIA VOR maintaining last assigned level or FL 90 whichever is higher. From 'BIA VOR, carryout published instrument approach procedure for assigned arrival runway. If higher, descend to FL90 over the published HOLD for the respective Instrument approach procedure and then carryout published instrument approach procedure for arrival runway.

3.1.6 PROCEDURE FOR NON-RNAV ARRIVALS

3.1.6.1 If RCF takes place at or beyond 50NM of BIA, continue on ATS route, (if given heading or flying offset, continue the last assigned heading/level for 3 minutes after detecting RCF or selecting code 7600 and Re-join ATS route) descending to cleared flight level or FL 90 whichever is higher. From 'BIA VOR', carryout published instrument approach procedure for arrival runway. If higher descend to FL90 over the published HOLD for the respective ILS approach and then carryout published instrument approach procedure for arrival runway.

3.1.6.2 If RCF takes place within 50NM of BIA, then aircraft shall maintain last assigned level and heading for 3 minutes after detecting RCF or selecting code 7600, whichever is later (If below FL90, climb and reach FL90 before crossing 25NM outbound from BIA VOR) and then proceed direct to BIA VOR maintaining last assigned level or FL90 whichever is higher. From 'BIA VOR', carryout published instrument approach procedure for arrival runway. If higher descend to FL90 over the published HOLD for the respective ILS approach and then carryout published instrument approach procedure for arrival runway.

4. PROCEDURE FOR DEPARTURE (BOTH RNAV AND NON RNAV)

4.1 DEPARTURE INTENDING TO CONTINUE TO DESTINATION

Any departing aircraft experiencing RCF and intending to continue to its filed plan destination shall:

4.1.1 Continue on assigned SID or heading climbing to or maintaining cleared level or FL90, whichever is higher.

4.1.2 Three minutes after setting Mode A/C code 7600 or reaching FL 90 or cleared Flight level (if higher than FL90) whichever is later:

a) If following 'SID', continue on SID to join ATS route and after joining ATS route climb to filed Flight level and continue as per the filed flight plan.

b) If following Radar heading, turn to join ATS route by shortest route maintaining cleared flight level/FL 90 whichever is higher. After joining ATS route, climb to filed Flight level and Continue as per the filed flight plan.

4.2 DEPARTURE INTENDING TO LAND BACK AT BENGALURU

Any departing aircraft experiencing RCF after departure and intending to land back at Kempegowda International Airport shall:

4.2.1 Continue on assigned SID or heading, climbing to or maintaining cleared level or FL90 whichever is higher.

4.2.2 Three minutes after setting Mode A/C code 7600 or reaching FL90 or cleared Flight level (if higher than FL90) whichever is later, take a turn to proceed direct to BIA VOR.

4.2.3 If required to burn fuel, then remain in the hold.

4.2.4 From BIA VOR carryout published instrument approach procedure for arrival runway. If higher descend to FL90 over the published HOLD for the respective ILS approach and then carryout published instrument approach procedure for arrival runway.

5 RUNWAY CAPACITY ENHANCEMENT MEASURES

5.1. INTRODUCTION:

Runway Occupancy Time (ROT) is one of the key determinants of the airport runway capacity and is especially critical for an airport which is experiencing a robust growth in air traffic in recent years. The objective is to lay down procedures aimed at reducing the ROT and enhancing runway utilization and capacity.

5.2. PROCEDURES:

i. All aircrafts shall make every effort to minimize the Runway Occupancy Time (ROT).

ii. No aircraft shall request for any direct routing. Direct routings shall be issued by controller only for the purpose of conflict resolution.

iii. During segregated mode of operations, runway changeover may cause a delay of 15 to 20 minutes for Arrival/Departure.

5.2.1 DEPARTURE PROCEDURE:

i. All departures requiring or able intersection departure shall inform Bengaluru Ground before requesting taxi.

ii. Taxiing aircraft should maintain a speed of not less than 15 knots on the straight portion of taxiways and between 8-12 knots during turning maneuvers during fair weather conditions.

iii. ATC may alter the departure sequence of an aircraft, which is not adhering to the procedure enumerated in para 5.2.1 (ii) above to optimize the runway utilization. ATC may also alter the departure sequence of an aircraft to facilitate maximum number of departures with the least average delay.

iv. Pilots shall complete all mandatory pre-departure checks before entering the active runway for departure so that the aircraft is in a position to take off immediately upon receipt of take-off clearance. Pilots not able to comply with this requirement must notify SMC/TWR before reaching the holding point.

v. When the aircraft is issued with a line-up and take-off clearance at the taxi holding point it shall be in a position to line up and initiate an immediate takeoff in one continuous movement. Pilots not able to comply with this requirement must notify TWR before lining up.

vi. When the aircraft is issued with a take-off clearance after lining up on the runway it shall commence take-off roll immediately.

vii. All departures shall contact Bengaluru Approach 127.75 MHZ after passing 3800 feet, unless instructed otherwise by the tower.

viii. All RNAV departures, on first contact with Bengaluru Approach shall transmit only their call sign, Altitude and assigned SID.

ix. All non-RNAV departures or departures not assigned any SID shall transmit their destination and first reporting point or route in place of SID.

x. Take-off Run Available from different intersections for RWY 09R/27L and RWY 09L/27R are available in charts under AD2.24

xi. All ATR/DH8D departure shall use A10/A4 Intersection departure when RWY in use 09L/27R respectively except when mandated by ACFT OPR requirements. However, ATC may advise them to use full length of RWY due to traffic considerations.

5.2.2 ARRIVAL PROCEDURE

i. All arrivals, on first contact with Bengaluru Approach shall transmit only their Call sign and level.

ii. The speed control measures as promulgated vide eAIP ENR 1.6 shall be strictly followed by flight crew unless exempted by ATC on request from the pilot.

iii. Pilots are reminded that rapid exit from the runway enables ATC to apply minimum spacing on final approach that will achieve maximum runway utilization and will minimize the occurrences of 'go around'.

iv. All arrivals, after vacating RWY, shall hold at first intermediate holding point (IHP), unless cleared to the contrary by appropriate ATC unit.

v. Arrivals need not inform ATC on reaching parking stand.

vi. Details of the locations of taxiways with respect to threshold for RWY 09R/27L and RWY 09L/27R are given in charts under AD2.24

vii. As all ARR aircraft to vacate RWY via preferred RET as mentioned below in table (5.3.1 & 5.3.2) or inform inability to comply, to Bengaluru APP at 20 NM from touchdown.

viii. All arriving aircraft after vacating the NORTH RUNWAY (09L/27R), to contact bengaluru ground on 121.650 MHZ.

ix. All arriving aircraft after vacating the SOUTH RUNWAY (09R/27L), to contact bengaluru ground on 121.775 MHZ.

5.3. PREFERRED EXIT TAXIWAYS:

5.3.1 The preferred exit taxiways for RWY 09R and 27L are as follows:

RWY Designation	Aircraft Type	Preferred Exit TWY	Distance from THR (M)
09R	Code A, B, C	H5 (RET)	1840
09R	Code D, E, F	H4 (RET)	2350
27L	Code A, B, C	H6 (RET)	1840
27L	Code D, E, F	H7 (RET)	2350

5.3.2 The preferred exit taxiways for RWY 09L and 27R are:

RWY Designation	Aircraft Type	Preferred Exit TWY	Distance from THR (M)
09L	Code A, B, C	A6 (RET)	1840
09L	Code D, E, F	A5 (RET)	2475
27R	Code A, B, C	A7 (RET)	1840
27R	Code D, E, F	A8 (RET)	2346

*Distance: The distances mentioned are the distances from the runway threshold to the RET turn off curve i.e. the point at which an aircraft starts to exit the RWY.

5.4. Details of TORA

Runway	Taxiway	TORA from Taxiway (Metres)

09L	A11	4000
	A10	3350
27R	A1	4000
	A2	3896
	A4	2700
09R	H10	4000
	H9	3896
	H8	3364
27L	H1	4000
	H2	3896
	H3	3287

VOBL AD 2.23 ADDITIONAL INFORMATION

1.DETAILS OF TAXIWAYS

S.No	Designator	Width (M)	PCR Strength	Surface
1	A	23	888/F/B/W/T	Asphalt
2	A1	23	1143/R/B/W/T 888/F/B/W/T	Concrete Asphalt
3	A2	23	888/F/B/W/T	Asphalt
4	A4	23	888/F/B/W/T	Asphalt
5	A5	23	888/F/B/W/T	Asphalt
6	A6	23	888/F/B/W/T	Asphalt
7	A7	23	888/F/B/W/T	Asphalt
8	A8	23	888/F/B/W/T	Asphalt
9	A9	23	888/F/B/W/T	Asphalt
10	A10	23	888/F/B/W/T	Asphalt
11	A11	23	1143/R/B/W/T 888/F/B/W/T	Concrete Asphalt
12	B	23	1069/F/C/W/T	Asphalt
13	B1	23	1069/F/C/W/T	Asphalt
14	B2	23	1069/F/C/W/T	Asphalt
15	B3	15	812/F/B/W/T	Asphalt
16	B4	23	888/F/B/W/T	Asphalt
17	B6	23	888/F/B/W/T	Asphalt
18	D	23	1069/F/C/W/T	Asphalt
19	F	23	1069/F/C/W/T	Asphalt
20	G	23	1069/F/C/W/T	Asphalt
21	G1	23	1069/F/C/W/T	Asphalt
22	G4	23	1069/F/C/W/T	Asphalt
23	H	23	1069/F/C/W/T	Asphalt
24	H1	23	1143/R/B/W/T 1069/F/C/W/T	Concrete Asphalt
25	H2	23	1143/R/B/W/T 1069/F/C/W/T	Concrete Asphalt
26	H3	23	1069/F/C/W/T	Asphalt
27	H4	23	1069/F/C/W/T	Asphalt
28	H5	23	1069/F/C/W/T	Asphalt
29	H6	23	1069/F/C/W/T	Asphalt
30	H7	23	1069/F/C/W/T	Asphalt

31	H8	23	1069/F/C/W/T	Asphalt
32	H9	23	1143/R/B/W/T 1069/F/C/W/T	Concrete Asphalt
33	H10	23	1143/R/B/W/T 1069/F/C/W/T	Concrete Asphalt
34	K	23	888/F/B/W/T	Asphalt
35	M	23	1069/F/C/W/T	Asphalt
36	N	23	1069/F/C/W/T	Asphalt
37	N1	23	1069/F/C/W/T	Asphalt
38	N2	23	1069/F/C/W/T	Asphalt
39	P	23	1069/F/C/W/T	Asphalt
40	P1	23	1069/F/C/W/T	Asphalt
41	Q	23	1069/F/C/W/T	Asphalt
42	R	23	1069/F/C/W/T	Asphalt
43	S	23	1069/F/C/W/T	Asphalt

2. RUNWAY HOLDING POSITIONS:			
S. No.	Description	Coordinates	Elev (FT) EGM-08
RWY HOLDING POSITION (09R/27L)			
1	RUNWAY HOLDING POSITION H1	131125.41N 0774336.36E	2960
2	RUNWAY HOLDING POSITION H2	131125.42N 0774333.33E	2960
3	RUNWAY HOLDING POSITION H3	131125.60N 0774312.58E	2960
4	RUNWAY HOLDING POSITION H4	131126.13N 0774253.71E	2959
5	RUNWAY HOLDING POSITION H5	131126.27N 0774236.78E	2960
6	RUNWAY HOLDING POSITION H6	131126.39N 0774223.95E	2961
7	RUNWAY HOLDING POSITION H7	131126.54N 0774207.00E	2963
8	RUNWAY HOLDING POSITION H8	131126.35N 0774145.53E	2966
9	RUNWAY HOLDING POSITION H9	131126.50N 0774127.39E	2968
10	RUNWAY HOLDING POSITION H10	131126.53N 0774124.37E	2969
RWY HOLDING POINT (09L/27R)			
1	RUNWAY HOLDING POSITION A1	131221.08N 0774322.22E	2917
2	RUNWAY HOLDING POSITION A2	131221.11N 0774319.20E	2919
3	RUNWAY HOLDING POSITION A4	131221.50N 0774242.44E	2942
4	RUNWAY HOLDING POSITION A5	131221.20N 0774238.78E	2944
5	RUNWAY HOLDING POSITION A6	131221.31N 0774222.63E	2956
6	RUNWAY HOLDING POSITION A7	131221.42N 0774209.86E	2966
7	RUNWAY HOLDING POSITION A8	131221.52N 0774157.91E	2977
8	RUNWAY HOLDING POSITION A9	131221.68N 0774139.66E	2986
9	RUNWAY HOLDING POSITION A10	131222.04N 0774131.41E	2989
10	RUNWAY HOLDING POSITION A11	131222.22N 0774110.24E	2999
11	DVOR CHECK POINT A-1	131220.76N 0774322.22E	2918
12	DVOR CHECK POINT A-11	131221.89N 0774110.24E	2998
13	MRO (HPER)	131204.68N 0774108.31E	2997
TAXIWAY INTERSECTION (09R/27L)			
1	RWY AND TAXIWAY-H1 INTERSECTION	131121.89N 0774336.33E	2964
2	RWY AND TAXIWAY-H2 INTERSECTION	131121.92N 0774333.30E	2964
3	RWY AND TAXIWAY-H3 INTERSECTION	131122.10N 0774312.55E	2964

4	RWY AND TAXIWAY-H4 INTERSECTION	131122.32N 0774246.85E	2964
5	RWY AND TAXIWAY-H5 INTERSECTION	131122.47N 0774229.91E	2966
6	RWY AND TAXIWAY-H6 INTERSECTION	131122.46N 0774230.75E	2966
7	RWY AND TAXIWAY-H7 INTERSECTION	131122.61N 0774213.82E	2967
8	RWY AND TAXIWAY-H8 INTERSECTION	131122.85N 0774145.50E	2971
9	RWY AND TAXIWAY-H9 INTERSECTION	131123.00N 0774127.36E	2973
10	RWY AND TAXIWAY-H10 INTERSECTION	131123.03N 0774124.34E	2973
RWY INTERSECTION (09L/27R)			
1	RWY AND TWY INTERSECTION A1	131224.65N 0774322.27E	2919
2	RWY AND TWY INTERSECTION A2	131224.68N 0774319.23E	2921
3	RWY AND TWY INTERSECTION A4	131224.99N 0774242.47E	2944
4	RWY AND TWY INTERSECTION A5	131225.08N 0774232.05E	2953
5	RWY AND TWY INTERSECTION A6	131225.22N 0774215.86E	2968
6	RWY AND TWY INTERSECTION A7	131225.21N 0774216.71E	2967
7	RWY AND TWY INTERSECTION A8	131225.32N 0774204.76E	2977
8	RWY AND TWY INTERSECTION A9	131225.47N 0774146.53E	2989
9	RWY AND TWY INTERSECTION A10	131225.60N 0774131.44E	2995
10	RWY AND TWY INTERSECTION A11	131225.79N 0774110.29E	3002

3. Distance of Taxiways from Runway Threshold:

FROM THRESHOLD	TAXIWAY	DISTANCE IN METER
27L	H2	104
27L	H6	1840
27L	H7	2350
27L	H8	3364
09R	H3	3287
09R	H4	2350
09R	H5	1840
09R	H9	104
27R	A2	104
27R	A7	1840
27R	A8	2346
27R	A9	2895
27R	A10	3350
09L	A4	2700
09L	A5	2475
09L	A6	1840

4. Ground Handling Arrangements:

All operators, operating to/from Kempegowda International Airport, Bengaluru shall employ the services of ground handling agent authorized by BIAL Operators should contact the Kempegowda International Airport, Bengaluru for further information on ground handling arrangements.

All Non-Scheduled operators operating to/from KIA, B to make necessary ground handling arrangements with GHA's in advance.

The three Ground Handling Agents are Air India-SATS, Globe Ground India and Menzies Aviation.

The contact details are as follows:

Air India SATS : +91-9538925272 / +91-9538086027

Globe Ground India : +91-9741393114 / +91-9740433990

Menzies Aviation: +91-9900543254/ +91-9187227652

5. Aircraft stands:

Refer Aircraft/ Parking Docking Chart for deatils of parking stands.

6. PARKING RESTRICTIONS ON STANDS:

STAND NO.	NO PARKING		STAND NO.	NO PARKING		STAND NO.	NO PARKING		STAND NO.	NO PARKING
V36	-		14	13, 15		51*	51W, 52W		83	82
V37	V38		15	14, 16*		52W	52*		83A	-
V38	V37, V39		16*	15, 17		52*	52W, 53W		84	85
V39	V38		17	18, 16*		53W	53*		84A	-
V40	V41		18	17, 19		53*	53W, 54W		85	84,86
V41	V40, V42		19	18		54W	53*		86	85
V42	V41		20	21		54	55*		86A	-
V43	V44		21	20, 22		55*	54, 56		87	-
V44	V43, V45		22	21		56	55*,57*		88	-
V45	V44		23	24		57*	56, 58		89	-
V46	V47		24	23, 25		58	57*		90	-
V47	V46, V48		25	24		59	60		91	-
V48	V47		26	27		60	59, 61		92	-
V49	V50		27	26, 28		61	60,62		93	-
V50	V49, V51		28	27		62	61, 63		94	-
V51	V50		29	30		63	62,64		95	-

V52	V53	30	29, 31	64	63, 65	96	-
V53	V52, V54	31	30	65	64	97	-
V54	V53	32	-	66	67	98	-
V55	-	33	34	67	66, 68	99	-
V56	V57	34	33, 35	68	67	100	-
V57	V56, V58	35	34	69	70	101	-
V58	V57	36	37	69A	-	102	-
V59	V60	37	36, 38	70	69,71	103	-
V60	V59, V61	38	37	71	70	104	-
V61	V60	39	40*	71A	-	105	-
V62***	-	40*	39, 41	72	73	106	-
V63	-	41	42*	72A	-	107	-
V68	V69	42*	41, 43	73	72,74	108	-
V69	V68,V70	43	-	74	73	109	-
V70	V69	45	-	74A	-	110	-
1	2	47	-	75	76	111	-
2**	1, 3	49	-	75A	-	112	-
3	2	501	502	76	75,77	113	114
4	5	502	501,503	77	76	114	113,115
5	4, 6	503	502	77A	-	115	114
6	5	504	505*	78	79	116	-
7	8	505*	504, 506	78A	-	117	-
8	7, 9	506	505*	79	78,80	118	119
9	8	507		80	79	119	118,120
10	11	508		80A	-	120	119
11	10, 12	509		81	82	121	122
12	11	510		81A	-	122	121,123
13	14	51W	51*	82	81,83	123	122

* If code D/E (wide body) type of aircraft parked on these stands adjacent stands to be kept vacant. If Code C type of aircraft is parked on these stands, no restriction on adjacent stands.

** If A380 is parked on stand 02, adjacent stands 01, 03 & 04 shall be kept vacant.

***If V63 is allocated to B747-8, stand V62 will be restricted to the aircraft having wingspan up to 64.0 m only.

7. Mode of Operations

Segregated Mode of Operations shall be used at Kempegowda International Airport.

Runway 27 Mode:

A segregated mode of operation with Runway 27L is used for arrivals and Runway 27R is used for departures.

Runway 09 Mode:

A segregated mode of operation with Runway 09L is used for landing and Runway 09R is used for departures.

In case of non-availability/Scheduled maintenance of either runway, mixed mode operations will be conducted on the other runway and such decision will be taken by the WSO concerned.

Classification of Apron

Apron-1: Stand 33A to 49; 501 to 506 & 510 to 507; 51W to 65

Apron-2: Stand V68 to 32 & Stand 66 to 93 & Stand 69A to 86A

Apron-3: Stand 94 to 106 & V55 to V63

Apron-4: Stand 107 to 123 & V36 to V54

Area of Control

- SMC-1 shall be responsible for pushback and taxi on Apron-1 & Apron-2.
- SMC-2 shall be responsible for pushback and taxi on Apron-3 & Apron-4.
- CLD shall change over departures from Apron-1 & Apron-2 to SMC-1 and departures from Apron-3 & Apron-4 to SMC -2 respectively.
- All arrival vacated from Northern Runway 09L/27R shall change over to SMC-1 and all arrivals vacated from Southern Runway 09R/27L shall change over to SMC-2.

8. General Guidelines for Pushback and Taxying

8.1 All stands are power-in and push back stands. Power out is not permitted from any stand except in circumstances mentioned in SOP for Aircraft without pushback facility.

8.2 Apron is served by multiple aircraft stand taxi lanes/ taxiways L1, L2, L3, L4, L5, B1, B2, B3, B4, K, C, D, M, N and N1.

8.3 Before commencing pushback, the WOA (Walk Out Assistant)/Head set operator should confirm from the pilot or the engineer in the cockpit the direction to face the nose of the aircraft & the taxi lane to which the aircraft should be pushed back.Appropriate communication of the same to the push back driver is the responsibility of the WOA/Head set operator.

8.4 At KIAB the stand lead lines are marked to facilitate the pushback operator to position the aircraft on the correct taxi lane from the concerned parking stands. The pushback operators are advised to strictly follow aircraft stand lead line to ensure that the aircraft is positioned on the correct taxi lane.

8.5 The pushback operator shall ensure that the aircraft is aligned on the designated taxi lane. It should not be at an angle to the taxi lane.

8.6 The pushback operator shall ensure that the pushback is completed at the point where the stand lead in line meets the apron taxilane. Meaning that on completion of the pushback the nose wheel of the aircraft will be at the point where the stand lead in line meets the apron Taxilane corresponding to the stand from which push back was commenced, unless otherwise instructed by ATC.

8.7 Ensuring correct push back on to correct taxi lane as per instructions received from ATC is the responsibility of Airline operator/GHA.

8.8 Additional precautions need to be taken during push back to align on specified taxi lane. The pushback operator shall check with the head set operator, the direction of the pushback and the taxi lane on to which the aircraft should be pushed back.

8.9 Power out is not permitted from any stand except non-schedule aircraft without tow bar and having wingspan less than 32M and length 36M.

8.10 All departing aircraft to re-confirm the taxilane with the push back operator on completion of pushback.

8.11 Clearance of aircraft on apron from ground support equipment, vehicular traffic and aircraft is the responsibility of Airside Operations through Ground Handling Agent and airlines.

8.12 At KIAB the following stands can accommodate multiple aircraft (one at a time only) which falls in different Codes.

Stand 16 : Code C/D/E

Stands 40, 42 45, 47 : Code A/B/C/D/E

Stand 49 : Code A/B/C/D/E/F

Stands 51, 52, 53, 55, 57 : Code A/B/C/D/E (Only A332 & A342 in Code-E)

Stands V55, V62 : Code C/D/E

Stands V63 : Code C/D/E/B748

8.13 Marshalling service will be provided by Ground Handling Agent for all stands where AVDGS is not provided.

8.14 Aircraft stand taxi lanes L1, L3 & L4 are suitable for taxiing by Code A/B/C type aircraft only.

8.15 Aircraft stand taxi lane L2 is suitable for taxiing by Code D/E aircraft only. Taxi lane L5 is suitable for taxiing by A/B/C/D/E aircraft for HPER & MRO only (without payload) (For Code-F refer Code-F procedures).

8.16 Taxi lane C is connected to taxilane L1, L2, D, B1, M & N.

8.17 Taxi lane D is connected to taxilane L2, L3, C, B1, M, N and TWY-P & Q.

8.18 Taxiway B3 is compatible for Code C aircraft only. No IHP marking is available on B3.

8.19 Taxilane C, TWYs B2, B6, M, N, N1, N2, F, R & S are compatible for taxi by aircraft up to Code E. All other taxiways including RETs are compatible for taxi by all aircraft up to Code-F.

8.20 Taxilane L5 is connected to taxiway B6.

8.21 Simultaneous push back and start up by Code A/B/C aircraft on aircraft stand taxi lane L1 and L3 is permitted.

8.22 Simultaneous pushback and start up by Code A/B/C aircraft on aircraft stand taxi lane L1 or L3 and Code D/E aircraft on L2 shall not be permitted unless specified longitudinal separation as mentioned in para 9 does exist.

8.23 Simultaneous parallel taxi is not permitted on aircraft stand taxi lanes L1 and L2 or L2 and L3.

8.24 Changing of lanes from L1 to L3 and vice versa is permitted for taxiing Code C (or smaller type) aircraft under the guidance of Follow-Me service.

8.25 For aircraft entering via taxiway B2 to stand 4/5, “TAXI VIA B2 TO STAND 4/5” only should be used in taxi clearances since there is no L3/L2 guideline.

8.26 Lane change from L1-L3 and vice versa is not permitted for taxing aircraft when RVR reduces below 200 m/ at the discretion of Airside Operations Manager.

8.27 If a Code C and below type of aircraft is holding at runway holding position A2/A4/A10 and H2/H3/H8/H9 a Code D/E/F aircraft shall not be permitted to taxi behind on taxiway A/H..

8.28 If a Code D and above type of aircraft is holding at runway holding position on A2/A4/A10 and H2/H3/H8/H9 NO aircraft shall be permitted to taxi behind on taxiway A/H.

8.29 An aircraft vacating runway via exit taxiways is not simultaneously clear of runway and parallel taxiway A/H.

8.30 Rapid Exit Taxiways (RETs) shall not be used for departures even though RGLs are provided.

8.31 RETs of opposite runway shall not be used for vacation of aircraft (for runway 09L - do not vacate via A7/A8/A9, runway 27R – do not vacate via A5/A6, runway 09R – do not vacate via H6/H7 and for runway 27L - do not vacate via H4/H5).

8.32 No aircraft shall be permitted to taxi on taxiway A/H abeam rapid exit taxiways when an arriving aircraft is using these taxiways for runway vacation. The aircraft taxiing on taxiway A/H shall be asked to hold short of the particular taxiway in such cases.

8.33 If a Code C or below type of aircraft is holding at IHP facing south on taxiway B4/K, a Code E or below aircraft is permitted to taxi behind on taxiway A.

8.34 If a Code D and above type of aircraft is holding at IHP facing south on taxiway B4/K, no aircraft shall be permitted to taxi behind on taxiway A.

8.35 IHP marking is not provided on TWY-B2 short of TWY-A, aircraft is expected to hold short of Taxiway B on Taxiway B2 only (IHP marking available).

8.36 If any aircraft is holding at IHP facing north short of Taxiway B on taxiway B1/M/N, taxiway B and taxilane C is clear for taxi for aircraft up to Code E.

8.37 If any aircraft is holding at IHP facing south short of taxilane-C on taxilane B1/M/N, taxilane C is clear for aircraft taxi up to code E and taxiway B is clear for taxi for aircraft up to Code F.

8.38 If any aircraft is holding at IHP facing west short of taxilane-N on taxilane N1, taxilane N is clear for aircraft taxi up to code E and taxiway P is clear for taxi for aircraft up to Code F.

8.39 IHP markings are provided on

i) Taxilane N1 short of taxilane N

ii) Taxilane M short of taxilane C

iii) Taxilane N short of taxilane C

iv) Taxilane B1 short of taxilane C

8.40 Simultaneous use of taxiways A4 and A5 is not permitted for any type of aircraft.

8.41 180 degrees turn on Apron/Taxiways is not permitted.

8.42 Special arrangements may be made by BIAL to facilitate aircraft taxiing out of its own power from a stand and parking at locations other than the published stands due to operational requirements. Such arrangements shall be communicated to SMC by Airside Operations Manager stating the restrictions imposed there upon.

8.43 180-degree turn may be allowed for Code A/B/C aircraft at the end of runway.

8.44 Aircraft from stands 69A to 86A shall be issued with taxi-out instruction only when there is no conflicting traffic on taxiway A, as there is no intermediate holding position at taxiway B3.

8.45 Taxiway B3 is not clear when arrival is holding on L4 between stands 77A, 78A, 80A, 81A.

8.46 Only one pushback east and west of taxiway B3 is permitted at a time. Aircraft on taxiway B3 is not clear of taxiway A.

8.47 Aircraft from stands 94 & 95 facing east or west shall pull/push abeam stand 96 only.

8.48 Code C aircraft pushed back facing west from stands V68 shall pull abeam stand V70 to keep taxilane L1 clear for taxing of aircraft from/to taxilane C.

8.49 SMC-1 shall advise SMC-2 if the aircraft is asked to hold abeam stand 96 for any reason.

8.50 SMC-2 shall instruct arriving aircraft to hold abeam stand V63 on taxilane D and abeam stand 96 on C and change over to SMC-1 for further taxi.

8.51 SMC-1 shall instruct departing aircraft on TWY B to hold short of taxiway B1 and change over to SMC-2 for further taxi.

8.52 Departure from apron-1 and apron-2, SMC-1 shall hold aircraft abeam stand 93 on L1/ stand V69/V70 on L2/L3 and change over to SMC-2 for further taxi.

8.53 SMC 2 shall coordinate with SMC 1 regarding any aircraft pushing back/holding/taxiing out on taxilane L1/C.

8.54 Jet Blast Clearance

8.54.1 Code letters A/B/C/D/E/F is as per definition of Aerodrome reference codes given in DGCA, CAR Section-4, Series B, Part-1 & Annex 14.

8.54.2 The distance from tail at which blast velocity reduces to 56 Kmph for breakaway thrust is as follows (AviPlan).

i) Code C : 61 m

ii) Code D, E, F (except A332) : 180 m

iii) A 332 : 248 m

8.54.3 Pushback procedures are laid down considering minimum jet blast clearance as mentioned above.

9. PUSHBACK RESTRICTIONS:

9.1 Minimum Separation of Stands Required for Simultaneous Pushback and Startup on Apron 1 & 2

9.1.1 For the same direction pushback and start up, the following minimum separation is applicable:

i) Between Code A/B/C & Code A/B/C: ±4 Stands

Note: Between stands 90 to 93 ±3 Stands

ii) Between Code D/E/F & Code D/E/F: ±6 Stands

iii) Between code A/B/C & code D/E/F

a) Code A/B/C in front of Code D/E/F: ±4 stands

b) Code D/E/F in front of Code A/B/C: ±6 stands

NOTE: Except for stands 59 to 65 (only one pushback is permitted in this section)

9.1.2 For opposite direction pushback and start up

i) Tail to Tail

a) Between Code A/B/C & Code A/B/C: ±4 Stands

b) Between Code D/E/F & Code A/B/C/D/E/F: ±6 Stands

ii) Nose to Nose (only for Code C or smaller type aircraft, Taxi out with follow- me service to change lane L1 to L3 or vice versa):

a) Between Code A/B/C & Code A/B/C aircraft: ±3 stands

NOTE: The above-mentioned separations are applicable for all the aircraft except A332 which requires 11 stand separation.

9.1.3 For Departure & Arrival

i) When arrival coming to a stand behind the aircraft pushing back

a) Code A/B/C pushback - Arrival Code A/B/C: ±3 stands

b) Code A/B/C pushback - Arrival Code D/E/F: ±4 stands

c) Code D/E/F pushback - Arrival Code A/B/C/D/E/F: ±5 stands

Note: Care should be taken not to permit arrival to stand with ‘Follow-Me’ service, behind an aircraft pushed back, if the aircraft which has pushed back has been given taxi clearance. This is to prevent the ‘Follow-Me’ vehicle from the Jet Blast of the aircraft taxiing out.

ii) When arrival coming to a stand in front of aircraft pushed back

a) Code A/B/C pushback – Arrival code A/B/C: ±2 stand

b) Code A/B/C pushback – Arrival code D/E/F: ±2 stands

c) Code D/E/F pushback – Arrival code A/B/C/D/E/F: ±2 stands

Note: Care should be taken not to permit simultaneous taxi in and push back when only the minimum separation is available. In such cases, arrival should be allowed to taxi into the stand only after the push back is completed.

9.1.4 Minimum separation of stands required for simultaneous push back (one behind another in same direction) without engine startup of aircraft in front:

i) Between Code A/B/C – Code A/B/C: ±3 stands

ii) Between Code D/E/F – Code A/B/C/D/E/F: ±4 stands

9.1.5 Aircraft in the front shall be given engine start up only after the separation mentioned in above clauses is achieved by pulling or pushing further the aircraft involved.

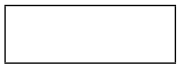
9.1.6 Minimum separation of stands required for simultaneous push back (tail to tail in opposite direction) without engine startup of aircraft involved:

i) Code A/B/C – Code A/B/C: ±5 stands

ii) Code D – Code A/B/C/D/E/F: ±5 stands

9.1.7 Aircraft involved shall be given engine start up only after the separation as mentioned above is achieved by pulling further the aircraft involved.

9.1.8 Stands 45, 47, 49, 502, 505 are southern stands and 51, 52, 53 are northern stands. Only one wide bodied aircraft shall pushback from stands 45 to 53 at a given time.



9.1.9 During pushback, stands with suffix 'W' will be considered as without suffix and five stand separations will be provided, for e.g., when aircraft is pushing back from stand 51 or 51W, the next aircraft can push back in the same direction from stand 56 only.

9.2 Minimum Separation of Stands Required for Simultaneous Pushback and Startup on APRON 3 & 4

9.2.1 For the same direction pushback and start up, the following minimum separation is applicable:

i) Between 94 & 106: ±3 Stands

ii) Between V55 & V63 (Including V53 & V54)

a) Between Code C & Code C ±3 Stands*

Note: (*2 stand separation for stands V61 & V63 and V54 & V56)

b) Between code C & code D/E/F

c) Code C in front of Code D/E/F ±3 stands

d) Code D/E/F in front of Code C/D/E/F ±6 stand

Code D/E pushed back from V53 facing east shall position at IHP marking (west of V54).

Note:

a) Wide body aircraft pushed back from stand V63 facing East shall pull abeam stand V62 for start- up due jet blast effect on equipment staging area.

b) Simultaneous push back from stand V68/V69 and V63 are clear of each other.

iii) Between V36 & V54

a) Between Code C & Code C: ±4 Stands

(Aircraft pushing back from stand V52, V53, V54 facing East on taxilane D shall push back at IHP marking west of V54).

Note: GHA's/Airlines shall ensure that the aircraft is positioned at IHP marking.

iv) Between code C & code D/E/F

a) Code C in front of Code D/E ±4 stands

b) Code D/E in front of Code C/D/E ±6 stands

Aircraft from stand V52, V53, V54 after pushback facing north on taxi lane M shall push abeam stand V51.

Note:

a) A332 from V53 shall be given facing east at IHP marking near V54 or west on taxilane D abeam stand 109.

b) If A332 from V53 pushing back facing north on M, 11 stands separation shall be provided.

v) Between 107 & 108

a) Only one pushback at a time.

vi) Between 109 & 115

a) Only one pushback at a time.

vii) Between 116 & 123

a) Between Code C & Code C ±4 Stands

b) Code C in front of Code D/E ±4 stands

c) Only one Code D/E push back at a time

9.3 Other Restriction On Pushbacks. Pushback Restrictions On Txl M (At The Extended Portion Of Txl N) Is As Flw :

1. No Pushback Shall Be Permitted From Stands V38, V39, V40 & V41 In Any Direction When Code C Acft Is Taxiing In/ out VIA M-N Or Vice Versa

2. In Case Code C Or Below Acft Is Pushed Back From Stands V38, V39, V40 & V41 In Any Direction, The Flw Push Back Procedures Will Be Flw To Permit Taxi Of Code C Acft On TXL M-TXL N And Vice Versa:
A) Code C Acft Pushing Back From Stand V40 Facing North/south To Pull/push Abeam Stand V42.
B) Code C Acft Pushing Back From Stand V39 Facing North To Push Deep Abm Stand V37.
3. Code C Acft Push Back From V37 Facing North And V42 Facing North/south Is Clear From TAXI IN/OUT VIA M-N/N-M.
4. Code D/e Acft Pushing Back From Stand V41 Facing North Shall Pull Abm Stand V44 To Clear Code C Acft Taxi In/out Via M-N/N-M.
Note: Only Code C Acft Are Permitted To Taxi Through The Extended Portion Of N To M-N/N-M To Mitigate The Jet Blast Impact Even Though Txl N Is Code-e Compliant)

10. PUSHBACK PROCEDURES

10.1 All Stands are power-in and push back.

10.2 APRON-1 and APRON-2 are served by multiple parallel aircraft stand taxi lanes L1, L2 and L3 (numbered from North to South). APRON-2 has additional taxi lane L4 which serves stands 69A to 86A only. APRON-3 is served by taxilane 'C' which connects to L1 & L2 and taxilane D which connects to L2 & L3. APRON-4 is connected to aircraft stand taxilane N, D and M.

10.3 Pushback from stands 69A to 86A (APRON-2) shall be on L4 only. Only one push back at a time shall be permitted from West of TWY-B3 & one push back from East of TWY-B3. All aircraft from stands 69A to 78A shall be given push back on L4 facing east and all aircraft from stand 80A to 86A shall be given push back facing west only , irrespective of runway in- use since aircraft can taxi out only via TWY B3.

10.4 Push back from APRON-3 shall be given on taxilane L1/C/D. (Stands 90 to 95 are connected to taxilane L1 & stands 96 to 106 are connected to taxilane C and stand V55 to V63 are connected to taxilane D)

10.5 All code D/E/F aircraft from APRON-1 and APRON-2 shall be given pushback on L2 only.

10.6 Push back from APRON-4 irrespective of aircraft type shall be

i) From stands 107, 108 & 116 to 123 on stand taxilane N

ii) From stands 109 to 115 on stand taxilane D.

iii) From stands V36 to V51 on taxilane M (from stands V36 to V42 shall pushback on taxilane M facing north only).

iv) From stands V52, V53 and V54 on taxilane D/M.

10.7 Simultaneous push back and start up by Code A/B/C aircraft on aircraft stand taxi lane L1 and L3 is permitted.

10.8 Simultaneous pushback and start up by Code A/B/C aircraft on aircraft stand taxi lane L1 or L3 and Code D/E/F aircraft on L2 shall not be permitted unless specified longitudinal separation as mentioned under push back restrictions.

10.9 When a Code C aircraft is given push back from stands 33/65 (or west of it) from APRON-1, another aircraft from stands 32/66 (or east of it) from APRON-2 can be given push back simultaneously.

10.10 If code D/E pushes back facing west from stands 34/64, push back from stand 30/67 in any direction is clear.

10.11 Other Restrictions

10.11.1 To keep B2 Clear

i) No pushback shall be permitted from stands 01, 02, 03, 04, 05, 06, 07, 08, 89 & 90 in any direction when aircraft is taxiing in/out via taxiway B2. If aircraft pushed back, the following restrictions apply:

ii) Code C aircraft from stand 06 & 07 shall push/pull abeam stand 09 facing east or west.

iii) Code D/E aircraft pushed back from stands 02, 05 & 08 shall push abeam stand V69 if facing east or west.

iv) Code D/E/F aircraft pushed back facing east from stand 02 shall pull abeam stand V69 and arrival to stand 03 shall not be permitted in this case.

v) Code C aircraft pushed back facing east or west from stands 01, 03 & 04 shall pull abeam stand V70.

vi) Code C aircraft pushed back from stand V68, V70, 01, 03 ,04, 06 and 07 is clear of another aircraft taxiing in via TWY B2 for L1.

vii) Code C aircraft taxiing on L3 abeam TWY B2 is clear of another Code C aircraft joining L1 from B2 and vice-versa.

viii) Aircraft taxi to stand 05 is not allowed if there is a pushback on stand 89 facing east/west and stand 88 facing west.

ix) Code C aircraft pushed back facing east from stand 90 shall pull abeam stand 91.

x) Code C aircraft pushed back facing west from stand 89 shall pull abeam stand 88.

10.11.2 To keep TWY B4 clear for aircraft taxi-in/out

i) Aircraft from stand 32/66 shall establish abeam 32/66 if facing west and abeam stand 31/67 if facing east.

ii) Aircraft from stand 33/34/65 shall establish abeam 33/34/65 if facing east and abeam stand 35/37/63 if facing west.

10.11.3 To keep TWY B1 clear for aircraft taxi-in/out

i) Pushback from stand 100 facing WEST shall pull abeam stand 99.

ii) Pushback from stand 101 facing EAST shall pull abeam stand 102.

iii) Aircraft pushing back from stand V56, V57, V58 facing East shall pull abeam stand V55.

iv) Aircraft pushing back from stand V57, V58 facing West shall push abeam stand V56.

v) Aircraft pushing back from stand V59, V60 facing East shall push abeam stand V61.

vi) Aircraft pushing back from stand V59, V60, V61 facing West shall pull abeam stand V62.

vii) When aircraft is pushing back from stand V56 to V61, aircraft joining D via B1 and vice versa is not clear.

10.11.4 To keep taxilane N1 clear:

i) Aircraft from stand V43, V44 after pushback facing south shall pull/push abeam stand V45.

ii) Aircraft from stand V43, V44, V45 after pushback facing north shall pull/push abeam stand V46.

iii) Aircraft from stand V41, V42 after pushback facing north shall push abeam stand V40.

iv) Aircraft pushing back from stand 123 facing north shall pull abeam stand 121.

v) Aircraft pushing back from stand 122 facing north shall pull abeam stand 119.

Note: Aircraft pushed back from stand V36 to pull abeam stand V37 to avoid jet blast on perimeter road.

10.11.5 Restriction for stands 106, 107 & 108

i) Pushback from stand 106 facing WEST shall pull abeam stand 105 to keep taxilane M clear.

ii) Pushback from stand 107 facing SOUTH shall pull abeam stand 108 to keep Taxiway B clear.

iii) Pushback from stand 108 facing north shall pull abeam stand 107 to keep taxilane C clear.

10.11.6 Restriction for stands 109 to 115

i) Pushback from stand 109 facing east shall pull abeam stand 110 to keep taxilane C and check N clear.

ii) Code D/E/F aircraft pushback from stand 114 facing west on taxilane D shall pull abeam stand 110 to keep TWY P clear.

iii) Pushback from stands 112/113 facing west on stand taxilane D shall pull abeam stands 114 to keep Taxiway P clear.

10.11.7 Restriction for stands 116 to 123

i) Pushback from stand 116 facing south shall pull abeam stand 117 to keep taxilane D clear.

ii) Pushback from stand 122 facing north shall pull abeam stand 119 to avoid jet blast effect on vehicular traffic operating on perimeter road.

10.11.8 Pushback restriction during IAPP (Isolated Aircraft Parking Position) activation

In the event of activation of IAPP, Tower controller will inform AOM, in turn AOM shall ensure the following:

i) Cross Service Road 1 is closed by placing the barricades on southern and northern side of CSR-1 & Entry to CSR-1 from Inner Service Road is also closed by placing barricades. During night, ensure that red glimp lamps are also placed for blocking the service roads.

ii) Barricades are placed east of Cross Service Road 1 (CSR 1) & at taxiway K and the taxi lanes between CSR-1 & taxiway K shall be closed to all traffic.

iii) Procedures listed out in SOP-35 shall be adhered to during IAPP activation.

iv) Aircraft from stand 33 to 506 & 507 to 65 shall push back facing east only. Facing west not approved as taxiway K is closed.

v) Narrow body aircraft on stand 506,505,507 & 508 after pushback on L3/L1 facing east shall be pulled abeam stand 504/509 for start-up before taxi. Wide body aircraft on stand 505 shall push back on L2 facing east and pull abeam stand 49 for start-up.

vi) Aircraft on stand 502 after pushback on L2 facing east shall be pulled abeam stand 49 for start-up and taxi.

10.11.9 Pushback restrictions on TXL M (at the extended portion of TXL N) is as follows:

a) No pushback shall be permitted from stands V38, V39, V40 & V41 in any direction when Code C aircraft is taxiing in/ out via M-N or vice-versa.

b) In case Code C or below aircraft is pushed back from stands V38, V39, V40 & V41 in any direction, the following push back procedures will be followed to permit taxi of Code C aircraft on TXL M-TXL N and vice versa:

i) Code C aircraft pushing back from stand V40 facing north/south to pull/push abeam stand V42.

ii) Code C aircraft pushing back from stand V39 facing north to push deep abeam stand V37.

c) Code C aircraft push back from V37 facing north and V42 facing north/south is clear from taxi in/out via M-N/N-M.

d) Code D/E aircraft pushing back from stand V41 facing north shall pull abm stand V44 to clear code c acft taxi in/out via M-N/N-M.

NOTE: Only Code C Aircraft are permitted to taxi through the extended portion of N to M-N/N-M to mitigate the jet blast impact even though TXL N is CODE-E compliant.

11. TAXIING OUT FROM APRON:

VIA Taxiway-B4

a. Code C or smaller type of aircraft taxiing towards west simultaneously on Taxi lane L1 and L3 for proceeding via Taxiway `B4' shall hold abeam stand 69 and 28 respectively to give way to aircraft in sequence.
b. All aircraft Taxiing towards west on L1, L2, L3 to turn right after crossing Stand Lead-in Lines for stand 66, 67 and 32 respectively for Taxiway 'B4'.
c. All aircraft Taxiing towards east on L1, L2, L3 to turn left after crossing Stand Lead-in Lines for stand, 65, 64 and 33 respectively for Taxiway 'B4'.

VIA Taxiway-K

a. All aircraft Taxiing towards west on L1, L2, L3 to turn right for Taxiway 'K'.

VIA Taxiway-B2

a. All aircraft Taxiing towards east on L1, L2, L3 to turn left after crossing Stand Lead-in Lines for stand 89, 05 and 06 respectively for Taxiway 'B2'.
b. All aircraft taxing towards west on L1, L2, L3 to turn right **a**fter crossing Stand Lead-in Lines for stand 90, 02 and 03 respectively for Taxiway 'B2'.

12. TAXIING INTO APRON:

12.1 VIA Taxiway-B4

i) All Code A/B/C type of aircraft taxiing for Southern stands V68 to 32 shall turn left on Aircraft Stand Taxi Lane "L3".

ii) All Code A/B/C type of aircraft taxiing for Southern stands 33 to 506 shall turn right on Aircraft Stand Taxi Lane "L3".

iii) All Code-A/B/C type of aircraft taxiing for Northern stands 65 to 507 shall turn right on Aircraft Stand Taxi Lane "L1".

iv) All Code-A/B/C type of aircraft taxiing for stands 66 to 123 shall turn left on Aircraft Stand Taxi Lane "L1/C" or L3/D.

v) All Code-D/E type of aircraft taxiing for stands V69 to 30 and 67 to 122 shall turn left on Aircraft Stand Taxi Lane “L2”.

vi) All Code-D/E type of aircraft taxiing for stands 34 to 502 and 51 to 64 shall turn right on Aircraft Stand Taxi Lane “L2”.

12.2 VIA Taxiway “K”

i) All Code-A/B/C type of aircraft taxiing for Northern stands shall turn left on Aircraft Stand Taxi Lane “L1”.

ii) All Code A/B/C type of aircraft taxiing for Southern stands shall turn left on Aircraft Stand Taxi Lane “L3”.

iii) All Code D/E/F type of aircraft taxiing to apron shall turn left to follow Aircraft Stand Taxi Lane “L2” for allotted stand.

12.3 VIA TWY-B3

i) Aircraft allotted Stands 69A, 71A, 72A, 74A, 75A, 77A, 78A (west of TWY B3) shall turn right after TWY-B3 on L4 and aircraft allotted stands 80A, 81A, 83A, 84A and 86A (east of TWY-B3) shall turn left after TWY-B3 on L4 and proceed to respective parking stands.

12.4 VIA TWY-B2

i) All Code A/B/C type of aircraft taxiing for Southern stands 06 to 506 shall turn right on Aircraft Stand Taxi Lane “L3”.

ii) All Code A/B/C type of aircraft taxiing for stand V68, V70, 01, 03 shall turn left on Taxi Lane ‘L3”.

iii) All aircraft proceeding for stands 04 and 05 to follow the appropriate stand lead in lines from Taxi way ‘B2’.

iv) All Code A/B/C type of aircraft taxiing for Northern stands 89 to 507 shall turn right on Aircraft Stand Taxi Lane “L1”.

v) All Code A/B/C type of aircraft taxiing for stands 90 to 123 shall turn left on Aircraft Stand Taxi Lane “L1/C” or L3/D.

vi) All Code D/E/F type of aircraft taxiing for stands V69 and 02 shall turn left on Taxi Lane ‘L2”.

vii) All Code D/E type of aircraft taxiing for stands 08 to 502 & 51 to 85 shall turn right on Aircraft Stand Taxi Lane “L2”.

iv) Aircraft from stand 33 to 506 & 507 to 65 shall push back facing east only. Facing west not approved as taxiway K is closed.

v) Narrow body aircraft on stand 506,505,507 & 508 after pushback on L3/L1 facing east shall be pulled abeam stand 504/509 for start-up before taxi. Wide body aircraft on stand 505 shall push back on L2 facing east and pull abeam stand 49 for start-up.

vi) Aircraft on stand 502 after pushback on L2 facing east shall be pulled abeam stand 49 for start-up and taxi.

10.11.9 Pushback restrictions on TXL M (at the extended portion of TXL N) is as follows:

a) No pushback shall be permitted from stands V38, V39, V40 & V41 in any direction when Code C aircraft is taxiing in/ out via M-N or vice-versa.

b) In case Code C or below aircraft is pushed back from stands V38, V39, V40 & V41 in any direction, the following push back procedures will be followed to permit taxi of Code C aircraft on TXL M-TXL N and vice versa:

i) Code C aircraft pushing back from stand V40 facing north/south to pull/push abeam stand V42.

ii) Code C aircraft pushing back from stand V39 facing north to push deep abeam stand V37.

c) Code C aircraft push back from V37 facing north and V42 facing north/south is clear from taxi in/out via M-N/N-M.

d) Code D/E aircraft pushing back from stand V41 facing north shall pull abm stand V44 to clear code c acft taxi in/out via M-N/N-M.

NOTE: Only Code C Aircraft are permitted to taxi through the extended portion of N to M-N/N-M to mitigate the jet blast impact even though TXL N is CODE-E compliant.

11. TAXIING OUT FROM APRON:

VIA Taxiway-B4

a. Code C or smaller type of aircraft taxiing towards west simultaneously on Taxi lane L1 and L3 for proceeding via Taxiway `B4' shall hold abeam stand 69 and 28 respectively to give way to aircraft in sequence.
b. All aircraft Taxiing towards west on L1, L2, L3 to turn right after crossing Stand Lead-in Lines for stand 66, 67 and 32 respectively for Taxiway 'B4'.
c. All aircraft Taxiing towards east on L1, L2, L3 to turn left after crossing Stand Lead-in Lines for stand, 65, 64 and 33 respectively for Taxiway 'B4'.

VIA Taxiway-K

a. All aircraft Taxiing towards west on L1, L2, L3 to turn right for Taxiway 'K'.

VIA Taxiway-B2

a. All aircraft Taxiing towards east on L1, L2, L3 to turn left after crossing Stand Lead-in Lines for stand 89, 05 and 06 respectively for Taxiway 'B2'.
b. All aircraft taxing towards west on L1, L2, L3 to turn right **a**fter crossing Stand Lead-in Lines for stand 90, 02 and 03 respectively for Taxiway 'B2'.

12. TAXIING INTO APRON:

12.1 VIA Taxiway-B4

i) All Code A/B/C type of aircraft taxiing for Southern stands V68 to 32 shall turn left on Aircraft Stand Taxi Lane "L3".

ii) All Code A/B/C type of aircraft taxiing for Southern stands 33 to 506 shall turn right on Aircraft Stand Taxi Lane "L3".

iii) All Code-A/B/C type of aircraft taxiing for Northern stands 65 to 507 shall turn right on Aircraft Stand Taxi Lane "L1".

iv) All Code-A/B/C type of aircraft taxiing for stands 66 to 123 shall turn left on Aircraft Stand Taxi Lane "L1/C" or L3/D.

13. Code-F Operating Procedures

a. Parking stand 02, 49, 114, V57 & V60 are Code-F compatible. (If A380 parked on stand 02, stand 04 shall be kept vacant).

b. Parking stand 45, 47 & V63 are B748 compliant (Limited Code-F).

c. ATC will inform AOCC regarding any diversion of Code F aircraft as and when the information is received.

d. Taxilane D & B1 are Code-F complaint.

e. Portion of L2 between stand 49 and TWY-K is compatible for Code-F.

f. Portion of L2 between TWY-B4 and 49 is compatible for B748 (limited Code-F).

g. Portion of L2 between D & stand 02 is compatible for Code-F.

h. Portion of TWY-B2 between TWY-A & TWY-B, is code F compliant.

14. Ground Handling Procedures For Code-F.

a. Both North RWY 09L/27R and south RWY 09R/27L are compatible for Code F ACFT OPS.

b. Code F stand including A380: V57, V60, 02, 49, 114

c. B748 stand (Limited Code F): V63, 45, 47

d. Any code D/E/F pushing back from stand 114 facing West to pull forward abeam stand 110.

e. Whenever D/E/F aircraft is given pushback facing West from stand 114, ATC to ensure no code E/F is allowed to taxi on P abeam D till aircraft pulled forward abeam stand 110.

f. Utmost care needs to be taken by TWR/SMC to prevent conflicts on Taxiway A and Taxilane D with respect to Code F arrival/departure.

v) All Code-D/E type of aircraft taxiing for stands V69 to 30 and 67 to 122 shall turn left on Aircraft Stand Taxi Lane “L2”.

vi) All Code-D/E type of aircraft taxiing for stands 34 to 502 and 51 to 64 shall turn right on Aircraft Stand Taxi Lane “L2”.

12.2 VIA Taxiway “K”

i) All Code-A/B/C type of aircraft taxiing for Northern stands shall turn left on Aircraft Stand Taxi Lane “L1”.

ii) All Code A/B/C type of aircraft taxiing for Southern stands shall turn left on Aircraft Stand Taxi Lane “L3”.

iii) All Code D/E/F type of aircraft taxiing to apron shall turn left to follow Aircraft Stand Taxi Lane “L2” for allotted stand.

12.3 VIA TWY-B3

i) Aircraft allotted Stands 69A, 71A, 72A, 74A, 75A, 77A, 78A (west of TWY B3) shall turn right after TWY-B3 on L4 and aircraft allotted stands 80A, 81A, 83A, 84A and 86A (east of TWY-B3) shall turn left after TWY-B3 on L4 and proceed to respective parking stands.

15. JET BLAST ABATEMENT PROCEDURES:

Jet Blast prone areas on Main Apron

Taxi Routes	Aircraft Position	Jet Blast Prone Area

Aircraft exiting Apron via Taxiway-B4	Taxiing – OUT	Stands 29, 30, 31, 32, 33, 34, 35, 36, 37, 38
Aircraft entering Apron via Taxiway-B4 and turning left for stands 02 to 30 and 67 to 85	Taxiing – IN	Stands 63, 64, 65
Aircraft entering Apron Via Taxiway-B4 and turning right for stands 34 to 49 and 51 to 64	Taxiing – IN	Stands 66, 67, 68
Aircraft exiting Apron via Taxiway-B2	Taxiing – OUT	Stands 01, 02, 03, 04, 05, 06
Aircraft entering Apron Via Taxiway-B2 for stand no. 02& 05	Taxiing – IN	Stands 87, 88, 89, 90, 91, 92
Aircraft entering Apron via Taxiway-B1	Taxiing – IN	Stands 99, 100, 101, 102
Aircraft exiting Apron Via Taxiway-B1	Taxiing - OUT	Stands V57, V58, V59, V60
Aircraft entering Apron via Taxiway-M	Taxiing – IN	Stands 105, 106, 108, 109, 110
Aircraft entering Apron Via Taxiway-N	Taxiing – IN	Stands 107, 108, 109, 110
Aircraft exiting Apron Via Taxiway-N	Taxiing – OUT	Stands 107, 108
Aircraft entering Apron via Taxiway-D for Taxilane-N	Taxiing – IN	Stands 108, 109, 110, 116, 117
Aircraft exiting Apron via Taxilane N1	Taxiing - OUT	Stands V41, V42, V43, V44
Aircraft taxing via L2-D	Taxiing	Stand V68, V69, V70

16. APRON AREA:

Code D/E/F type of aircraft are not permitted to stop while Taxiing on these Jet Blast Prone Areas to avoid jet blast damage to vehicles and aircraft parked on the above-mentioned stands. If for any reason B747 type of aircraft stops on these Jet Blast Prone Areas further taxi should be continued only when the affected stands are vacant and ground movement is regulated. All jet Blast Prone Areas are well marked with apron markings to provide adequate information for all vehicles.

17. ARRIVING AIRCRAFT:

Aircraft can normally vacate the runway via designated exit taxiways:

Runway 09L– Taxiway A6, A5, A4, A2, A1

Runway 27R – Taxiway A7, A8, A9, A10, A11

Runway 09R– Taxiway H5, H4, H3, H2, H1

Runway 27L – Taxiway H6, H7, H8, H9, H10

18.DEPARTING AIRCRAFT

Aircraft can normally enter the runway for departure via the following taxiways:

Runway 09L – Taxiway A11, A10

Runway 27R – Taxiway A1, A2, A4

Runway 09R – Taxiway H8, H9, H10

Runway 27L – Taxiway H1, H2, H3

19. ASMGCS commissioned to enhance visual surveillance.

20. All aircraft with non ADS-B or unserviceable ADS-B equipment on board shall positively inform ATC on first contact during Arrival and before Pushback.

21. COMMISSIONING OF ADS-B

FAC : ADSB

FREQ : 1090MHZ

COORD : 131236.32N0774216.95E

SITE ELEV : 908M AMSL

STANDARD AIRPORT TAXI ROUTES (SATR)

1.Standard Airport Taxi Routes (SATR) is operational at KEMPEGOWDA INTERNATIONAL AIRPORT, BENGALURU (VOBL) to enhance ground control operations for both arriving and departing traffic. The purpose of Standard Airport Taxi Routes (SATR) is to use standard taxi routes instead of long and ambiguous taxi instructions. This alleviates R/T congestion and will thus allow controller to have better surveillance of the traffic in peak traffic times. The implementation of SATR will lead to reduction in workload of both ATCOs and Pilots.

2.A single SATR comprises of a number of taxiways which together form complete taxi instructions to a departure/arrival. SATR have been designed for RWY27L/09R operations (South RWY only). SATR comprises of taxiways outside the Apron

2.1 ARRIVAL SATR: The arrival SATR starts after the exit taxiway of Runway and terminates at the entry to the APRON. Further Taxiing for arriving aircraft will be advised by the ATC as per the designated parking stand

2.2 DEPARTURE SATR: The departure SATR starts from outside the APRON and terminates at the holding point at the beginning of specific Runway. The departing aircraft will join the SATR via Taxilane/Taxiway as instructed by the ATC.

2.3 A total of 4 Standard Airport Taxi Routes (SATR) are designed:

ROUTE 5	CONTINUE ON H - LEFT ON Q - LEFT ON A- HOLD SHORT OF B1.	RWY 09R ARR
ROUTE 10	RIGHT ON P- RIGHT ON G- LEFT ON G4 – RIGHT ON H-LEFT ON H10, HOLDING POINT RWY09R.	RWY 09R DEP
ROUTE 6	RIGHT ON H – LEFT ON G4 – RIGHT ON G- LEFT ON P-HOLD SHORT OF D	RWY 27L ARR
ROUTE 1	RIGHT ON Q – LEFT ON H-RIGHT ON H1, HOLDING POINT RWY27L	RWY 27L DEP

3.

OPERATING PRINCIPLE:

3.1Whenever SATR is not available due to closure or traffic, controller shall issue progressive taxi instructions

3.2Controller has the flexibility to issue alternate taxi instructions other than SATR to resolve/expedite traffic.

3.3All Twy Hold Short/XNG Instructions Shall Be Transmitted In Addn To STD AP TAX RTER Instructions As And When Required.

3.4ATCOs should provide taxi instructions to join SATR for departures. For arrivals taxi instructions should be given at the end of SATR up to parking bay.

3.5Pilots shall exercise Caution for the turns due to the complexity of taxi routes.

3.6If for any reason, the aircraft crew is unable to comply or understand the SATR instructions, timely request should be made for progressive taxi instructions.

3.7In case, ATCO cancels SATR for any aircraft, R/T instruction “CANCEL [SATR NAME], REVISED TAXI VIA... (REVISED ROUTING).” should be used.

3.8 STD AP TAX Routes Are Designed For First Entry RWY HLDG Point (Full Len Dep Like H10 FOR RWY09R AND H1 FOR RWY27L).However, STD AP TAX RTE Can Be Used For Intersection Dep Also By Adding The Entry TWY At The End Of The STD AP TAX RTE INSTRUCTION as mentioned in the example 4.1

3.9An aircraft may be advised to join a SATR via an intermediate taxiway. Such instruction will be given by defining the taxiway as a prefix to SATR.

3.10

As all the taxiways mentioned in SATR lies under the jurisdiction of SMC 2; Tower controllers and SMC 1 controller shall not use SATR.

Ex: ABC123 Bengaluru Ground: Cancel Route 1, Revised taxi via P-G-G1-H1 holding point RWY 27L.

Example: “Taxi to holding point Runway 09R via D - ROUTE 10” (means Taxi via D-then Follow SATR ROUTE 10 to holding point H10 RWY09R.)

4.PHRASEOLOGY:

4.1For Departures

(Aircraft Call sign) BENGALURU GROUND: TAXI TO HOLDING POINT RUNWAY [Designator] Via [taxiway joining respective parking bay to SATR] [SATR] Time [time]

Ex – ABC231 BENGALURU GROUND: TAXI TO HOLDING POINT RWY 27L VIA ROUTE 1.

Ex – XYZ509 BENGALURU GROUND: CONTINUE TAXI TO HOLDING POINT RWY 09R VIA ROUTE 10. (In case aircraft was instructed to hold short of any intermediate point while taxiing on Route 10).

Ex – PQR810 BENGALURU GROUND: TAXI TO HOLDING POINT RWY09R VIA ROUTE 10 H8. (In case of intersection departure from H8).

4.2For Arrivals

(Aircraft Call sign) BENGALURU GROUND: TAXI VIA [SATR] / TAXI VIA [SATR] HOLD SHORT OF [TWY]

Ex – LMN119 BENGALURU GROUND: TAXI VIA ROUTE 5.

Ex - RST194 TAXI VIA ROUTE 5, HOLD SHORT OF N2.

5.Refer SATR charts for the details.

VOBL AD 2.24 CHARTS RELATED TO AN AERODROME

1 Aerodrome Chart (07 August 2025)
2 Aerodrome Chart Hot Spot (07 August 2025)
3 Aircraft Parking/Docking Chart (08 August 2024)
4 Aerodrome Obstacle Chart-Type A (Operating Limitations) RWY 09L/27R (23 January 2025)
5 Aerodrome Obstacle Chart-Type A (Operating Limitations) RWY 09R (23 January 2025)
6 Aerodrome Obstacle Chart-Type A (Operating Limitations) RWY 27L (23 January 2025)
7 Aerodrome Obstacle Chart – Type B (23 January 2025)
8 Precision Approach Terrain Chart RWY 09
9 Precision Approach Terrain Chart RWY 27
10 Precision Approach Terrain Chart RWY 09R
11 Precision Approach Terrain Chart RWY 27L
12 ILS Critical And Sensitive Area - RWY 09R/27L
13 Take Off Run Available from Intersection RWY 09R (22 January 2026)
14 Take Off Run Available from Intersection RWY 27L
15 Take Off Run Available from Intersection RWY 09L
16 Take Off Run Available from Intersection RWY 27R
17 Location of Different Exit TWYs RWY 09R
18 Location of Different Exit TWYs RWY 27L
19 Location of Different Exit TWYs RWY 09L
20 Location of Different Exit TWYs RWY 27R
21 ILS or LOC Procedure RWY 09L (27 November 2025)
22 ILS Procedure RWY 27L
23 ILS or LOC Procedure RWY 27R (27 November 2025)
24 ILS CAT II/IIIA/IIIB RWY 27L
25 ILS or LOC Procedure RWY 09R (17 April 2025)
26 ATC Surveillance Minimum Altitude Chart
27 ILS Z Procedure RWY 09R (27 November 2025)
28 Coding Table 1- ILS Z Procedure RWY 09R (27 November 2025)
29 Coding Table 2- ILS Z Procedure RWY 09R (27 November 2025)
30 Standard Instrument Departure RWY 27L-RNAV1 (GNSS)
31 Standard Instrument Departure RWY 27L-RNAV1 (GNSS)-Coding Table 1
32 Standard Instrument Departure RWY 27L-RNAV1 (GNSS)-Coding Table 2
33 Standard Instrument Departure RWY 27L-RNAV1 (GNSS)-Coding Table 3
34 Standard Instrument Departure RWY 27L-RNAV1 (GNSS)-Coding Table 4
35 Standard Instrument Departure RWY 27L-RNAV1 (GNSS)-Coding Table 5
36 Standard Instrument Departure RWY 27L-RNAV1 (GNSS)-Waypoints
37 Standard Instrument Departure RWY 27R-RNAV1 (GNSS)
38 Standard Instrument Departure RWY 27R-RNAV1 (GNSS)-Coding Table 1
39 Standard Instrument Departure RWY 27R-RNAV1 (GNSS)-Coding Table 2
40 Standard Instrument Departure RWY 27R-RNAV1 (GNSS)-Coding Table 3
41 Standard Instrument Departure RWY 27R-RNAV1 (GNSS)-Coding Table 4
42 Standard Instrument Departure RWY 27R-RNAV1 (GNSS)-Coding Table 5
43 Standard Instrument Departure RWY 27R-RNAV1 (GNSS)-Waypoints

44 Standard Instrument Arrival RWY 09L-09R-RNAV1 (GNSS)

45 Standard Instrument Arrival RWY 09L-09R-RNAV1 (GNSS)-Coding Table 1

46 Standard Instrument Arrival RWY 09L-09R-RNAV1 (GNSS)-Coding Table 2

47 Standard Instrument Arrival RWY 09L-09R-RNAV1 (GNSS)-Coding Table 3

48 Standard Instrument Arrival RWY 09L-09R-RNAV1 (GNSS)-Coding Table 4

49 Standard Instrument Arrival RWY 09L-09R-RNAV1 (GNSS)-Waypoints 1

50 Standard Instrument Arrival RWY 09L-09R-RNAV1 (GNSS)-Waypoints 2

51 Standard Instrument Arrival RWY 27L-27R-RNAV1 (GNSS)

52 Standard Instrument Arrival RWY 27L-27R-RNAV1 (GNSS)-Coding Table 1

53 Standard Instrument Arrival RWY 27L-27R-RNAV1 (GNSS)-Coding Table 2

54 Standard Instrument Arrival RWY 27L-27R-RNAV1 (GNSS)-Coding Table 3

55 Standard Instrument Arrival RWY 27L-27R-RNAV1 (GNSS)-Coding Table 4

56 Standard Instrument Arrival RWY 27L-27R-RNAV1 (GNSS)-Coding Table 5

57 Standard Instrument Arrival RWY 27L-27R-RNAV1 (GNSS)-Waypoints 1

58 Standard Instrument Arrival RWY 27L-27R-RNAV1 (GNSS)-Waypoints 2

59 Standard Instrument Departure RWY 09L-RNAV1 (GNSS), GUNIM 8A, PEXEG 8A,TULNA 8A, VEMBO 8A (17 April 2025)

60 Standard Instrument Departure RWY 09L-RNAV1 (GNSS), GUNIM 8A, PEXEG 8A,TULNA 8A, VEMBO 8A-Coding Table 1 (17 April 2025)

61 Standard Instrument Departure RWY 09L-RNAV1 (GNSS), GUNIM 8A, PEXEG 8A,TULNA 8A, VEMBO 8A-Coding Table 2 (17 April 2025)

62 Standard Instrument Departure RWY 09L-RNAV1 (GNSS), AKTIM 8A, VAGPU 8A, OPAMO 8A,LATID 8A, SAI 8A, MSR 8A ,ANIRO 8A (17 April 2025)

63 Standard Instrument Departure RWY 09L-RNAV1 (GNSS), AKTIM 8A, VAGPU 8A, OPAMO 8A,LATID 8A, SAI 8A, MSR 8A ,ANIRO 8A-Coding Table 1(17 April 2025)

64 Standard Instrument Departure RWY 09L-RNAV1 (GNSS), AKTIM 8A, VAGPU 8A, OPAMO 8A,LATID 8A, SAI 8A, MSR 8A ,ANIRO 8A-Coding Table 2 (17 April 2025)

65 Standard Instrument Departure RWY 09L-RNAV1 (GNSS), AKTIM 8A, VAGPU 8A, OPAMO 8A,LATID 8A, SAI 8A, MSR 8A ,ANIRO 8A-Coding Table 3 (17 April 2025)

66 Standard Instrument Departure RWY 09R-RNAV1 (GNSS), AKTIM 8C, ANIRO 8C, GUNIM 8C, VAGPU 8C, OPAMO 8C, PEXEG 8C, TULNA 8C, VEMBO 8C, LATID 8C SAI 8C, MSR 8C (17 April 2025)

67 Standard Instrument Departure RWY 09R-RNAV1 (GNSS), AKTIM 8C, ANIRO 8C, GUNIM 8C, VAGPU 8C, OPAMO 8C, PEXEG 8C, TULNA 8C, VEMBO 8C, LATID 8C- SAI 8C, MSR 8C-Coding Table 1 (17 April 2025)

68 Standard Instrument Departure RWY 09R-RNAV1 (GNSS), AKTIM 8C, ANIRO 8C, GUNIM 8C, VAGPU 8C, OPAMO 8C, PEXEG 8C, TULNA 8C, VEMBO 8C, LATID 8C-SAI 8C, MSR 8C-Coding Table 2 (17 April 2025)

69 Standard Instrument Departure RWY 09R-RNAV1 (GNSS), AKTIM 8C, ANIRO 8C, GUNIM 8C, VAGPU 8C, OPAMO 8C, PEXEG 8C, TULNA 8C, VEMBO 8C, LATID 8C-SAI 8C, MSR 8C-Coding Table 3 (17 April 2025)

70 RNP Y Procedure RWY 09R (27 November 2025)

71 Coding Table - RNP Y Procedure RWY 09R (27 November 2025)

72 RNP Y Procedure RWY 27L

73 Coding Table - RNP Y Procedure RWY 27L

74 RNP Y Procedure RWY 27R

75 Coding Table - RNP Y Procedure RWY 27R

76 SMR Coverage Restriction Diagram

77 RNP Procedure RWY 09L

78 Coding Table - RNP Procedure RWY 09L

79 Standard Airport Taxi Route(STAR) Route 1(02 OCT 2025)

80 Standard Airport Taxi Route(STAR) Route 5(02 OCT 2025)

81 Standard Airport Taxi Route(STAR) Route 6(02 OCT 2025)

82 Standard Airport Taxi Route(STAR) Route 10(02 OCT 2025)

83 Taxi Route Chart-B747-8/8F (16 April 2026)

84 Taxi Route Chart - CODE F (A380/AN124/B748) (16 April 2026)